H-HANK: A Heteroskedastic Heterogeneous-Agent New Keynesian Model *

Roberto Colarieti
Harvard University

Tommaso Monacelli
Bocconi University, IGIER and CEPR

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Abstract

We lay out a *heteroskedastic* New Keynesian model, consistent with the evidence of cross-sectional variation in individual income risk in micro data. We obtain three main results. First, despite the assumption of CARA-Normal utility, marginal propensities to consume (MPCs) are heterogeneous across the income distribution, in line with the data. Second, the response of aggregate output to demand shocks hinges crucially on how the individual sensitivity of risk to income co-varies with the degree of individual income cyclicality across the income distribution. Third, the general equilibrium effects of monetary and fiscal policy can be suitably summarized by a set of observable cross-sectional sufficient statistics. Depending on the sign of those statistics, income heteroskedasticity may: (i) dampen or amplify the response of output to demand shocks; (ii) affect the local determinacy of the equilibrium; (iii) attenuate or exacerbate the forward guidance puzzle. Using PSID data we estimate that agents whose income is more sensitive to aggregate cyclical conditions are also more exposed to changes in risk, thereby leading to an overall amplification of demand shocks.

*JEL codes E52, E32, E21. Correspondence: robertocolarieti@fas.harvard.edu, tommaso.monacelli@unibocconi.it. We thank Edouard Challe, Pierfrancesco Mei, Francesco Nuzzi, and Gianluca Violante for insightful comments. All errors are our own.

1 Introduction

Traditionally, the literature on heterogeneous-agent models has been centered on the assumption that individual households face a homoskedastic income process: that is, the variance of the income process is homogeneous in the cross-section and constant over time. The assumption of homoskedasticity is also pervasive in the new wave of New Keynesian models with heterogeneous agents (so called HANK models). That assumption, however, is at odds with the evidence from micro data that points to significant cross-sectional *heteroskedasticity*: in particular, the evidence suggests that individual income risk is tied to *past* realizations of individual income (Guvenen et al., 2021; Arellano et al., 2021).

In this work, we explore the implications of heteroskedasticity in individual income by means of a CARA (constant-absolute-risk-aversion) utility model with a Gaussian heteroskedastic income process, building on the work by Caballero (1990) and, more recently, Acharya and Dogra (2020). Relative to the special case of homoskedasticity, our framework is able to account *both* for idiosyncratic risk and heterogeneity in marginal propensities to consume (MPCs) in an analytically tractable way, making the general equilibrium interaction between risk and the cross-sectional distribution of income extremely transparent.

Recent papers have shown that the *cyclicality* of income risk, i.e., how individual risk behaves during expansions and contractions, can be a source of either amplification (in the case of counter-cyclical risk) or dampening (in the case of pro-cyclical risk) of aggregate fluctuations.[1] For instance, agents react to negative aggregate demand shocks by decreasing their consumption relatively more when risk is counter-cyclical, since aggregate contractions are also associated with higher individual risk and precautionary savings. While this notion of risk-cyclicality captures the idea that idiosyncratic income risk may be *on average* higher during recessions,[2] it disregards the evidence that risk varies *across the earnings distribution*. Within our setting, realized changes in earnings endogenously affect the risk faced by each agent, and thus their optimal consumption and savings decisions. Suppose, consistent with our empirical evidence, that the variance of the individual income process is an inverse (and non-linear) function of realized earnings. In this case, a negative realization of income also increases individual risk, thereby strengthening at the margin the agent's motive for precautionary savings. As a result, the *sensitivity* of precautionary savings to income realizations will be heterogeneous across individuals.

We employ our heteroskedastic NK framework (H-HANK) to study three questions that have been at the core of the recent HANK literature: (a) the transmission of monetary and fiscal shocks; (b) equilibrium determinacy; (c) the so-called forward guidance puzzle (henceforth FGP).

We reach three main results. First, and unlike the corresponding homoskedastic case, the CARA-Gaussian setup augmented with heteroskedasticity features a non-degenerate *distribution* of

[1] For instance, Acharya and Dogra (2020) and Bilbiie (2018) for income risk, Ravn and Sterk (2021) for unemployment risk.

[2] See, e.g., Storesletten et al. (2004).

MPCs out of transitory income. This is due to realized earnings affecting income risk at the margin, and therefore also precautionary savings and consumption choices, as long as the marginal change in volatility varies over the earnings distribution. In particular, and conditional on the variance being a convex function of income, our model is able to generate MPCs which are decreasing over the income distribution, in line with the empirical evidence.[3] To gain intuition for this result, suppose that income volatility is decreasing and convex in realized earnings. Then any additional unit of income determines a decline in precautionary savings that is, at the margin, stronger for low-income households, who in turn react by spending more of the income shock. As a result, they exhibit relatively higher MPCs.

Second, the presence of heteroskedasticity has general equilibrium implications that are absent both in a representative-agent (RA) model and in a corresponding version of our framework with homoskedasticity. The interaction between income risk and the income distribution determines *indirect effects* that shape the responses of key aggregate variables to demand shocks, the determinacy properties of the equilibrium, and the FGP.

Third, and more specifically, the general equilibrium indirect effects of heteroskedasticity can be conveniently summarized in terms of three *sufficient statistics*: (1) the expected marginal change in income volatility out of realized individual earnings, capturing an *individual risk* effect; (2) the average sensitivity of individual income to aggregate income, capturing an *income cyclicality* effect; (3) the covariance between the marginal change in income volatility and the sensitivity of individual income to aggregate income, capturing a *risk-cyclicality covariance* effect .

The sufficient statistics described above allow to reduce the dimensionality of the model when describing the dynamic response of output to demand shocks. In particular, statistics (1) and (2) jointly capture how, *on average*, risk varies as a function of individual income in response to variations in aggregate output. Statistic (3), on the other hand, captures the extent to which individual risk *varies differently across agents* depending on how individual income is affected by aggregate income fluctuations.

Depending on the signs of these three sufficient statistics, heteroskedasticity could either amplify or dampen the response of output to demand shocks, affect the local determinacy of the equilibrium, and can either attenuate or exacerbate the FGP. In particular, negative signs for (1) and (3) (the most realistic case) amplify the effect of demand shocks on output, make the Taylor principle insufficient for local determinacy, and exacerbate the FGP, while for positive values of (1) and (3) the opposite occurs (i.e., there is dampening of demand shocks, the Taylor principle is no longer a necessary condition for equilibrium determinacy, and the FGP is attenuated).[4] Conversely, both in the benchmark complete markets RA model and in the homoskedastic version of our CARA

[3] Jappelli and Pistaferri (2014) estimate (using survey data from the Bank of Italy Survey of Household Income and Wealth) that annual MPCs are 11 percent higher in the lowest quintile of the income distribution relative to the highest quintile. Johnson et al. (2006) also find (with CEX data) that MPCs out of the 2001 U.S. tax rebates are higher for low-income households.

[4] In general, statistic (2) – the average sensitivity of individual to aggregate income – is positive, since average individual income is generally increasing in output.

framework, statistics (1) and (3) are constant and equal to zero at all times.

Notice that statistics (1) and (2) jointly describe the size of the effect on output of *average* changes in risk. For instance, a stronger average increase in income volatility generates a larger increase in precautionary savings, and therefore a larger contraction in consumption following a negative demand shock. Statistic (3), on the other hand, captures the *interaction* between income risk and income redistribution over the business cycle. Suppose that a negative demand shock has a first-round *direct* contractionary effect on output. To start with, lower aggregate income affects each agent depending on the heterogeneous sensitivity of individual income to aggregate income. In our heteroskedastic setup, however, any change in individual earnings affects also the volatility of the income process, and therefore saving and consumption decisions. If the covariance term (3) is (realistically) negative, those agents whose income falls relatively more out of a contraction in aggregate output are also the ones who experience a marginally stronger increase in idiosyncratic risk. At the margin, the same agents exhibit a relatively stronger precautionary saving motive and therefore will decrease their consumption relatively more. In other words, those agents whose income is more heavily affected by the contraction in aggregate output are also the ones who have a stronger incentive to decrease their consumption. This indirect *propagation through-risk* channel generates an amplification of the primitive shock on aggregate output.

As for equilibrium determinacy, a negative sign of the risk-cyclicality covariance (3), like in the previous example, makes the Taylor principle insufficient for local determinacy and requires monetary policy to react more strongly to a demand shock to stabilize the economy. On the contrary, a positive sign for the covariance (3) makes the equilibrium determinate also under an interest rate peg.[5]

Furthermore, a negative covariance (3) exacerbates the FGP. This results from the interaction of two mechanisms. For one, with a negative covariance, the agents expect that a monetary easing T periods ahead has an amplified positive effect on output T periods ahead (as previously described). In addition, agents expect higher output T periods ahead to cause output in $T-1$ to increase more than one-to-one. Overall, news about higher future output are compounded backward, resulting in a stronger increase in current output the longer the horizon T of the monetary easing. Conversely, a positive covariance (3) attenuates the FGP since these mechanisms operate in the opposite direction (i.e., future news are discounted).

In a subsequent step, we *estimate* both the size and the sign of the sufficient statistics described above. Using PSID data, we find that the variance of the unpredictable component of the income process is decreasing over the distribution of lagged income.[6] Therefore, the *individual risk effect* (i.e., statistic (1) above) – capturing how individual risk varies, on average, with lagged income – is negative in the data. Moreover, the marginal change in income volatility is larger (in absolute value) for poorer agents. Hence, the variance of the income process is a decreasing and convex function of lagged income, so that the individual risk effect is more negative for households in the

[5] Unlike the well-known result by Sargent and Wallace (1975).

[6] This finding is consistent with the literature, e.g., Arellano et al. (2021), Guvenen et al. (2021).

bottom part of the income distribution. Conditional on this pattern of the individual risk effect, our model can reproduce the empirical finding that poorer agents feature higher MPCs.

When focusing on individual earnings sensitivities, we find that agents in the bottom part of the income distribution display a more cyclical income (i.e., a stronger sensitivity of individual income to aggregate income).[7] Joint with the pattern observed for the individual risk effect, this implies that the risk-cyclicality covariance statistic (3) has a negative sign. In other words, those agents whose individual income is more exposed to aggregate income fluctuations also experience a stronger marginal change in their income volatility (i.e., they feature a "more negative" individual risk effect). Overall, a negative sign for both the average individual risk effect – statistic (1) – and the risk-cyclicality effect – statistic (3) – leads to an amplification of demand shocks, to the insufficiency of the Taylor principle for equilibrium determinacy, and to an exacerbation of the FGP.

1.1 Relation to the literature

Our work relates to several strands of the literature. Concerning the propagation of monetary and fiscal policy, Kaplan et al. (2018) disentangle *direct* and *indirect* effects of monetary policy on output and show that, in a HANK model, policy transmission works mainly through the indirect channel. We highlight a new general equilibrium channel of transmission of monetary and fiscal policy, stemming from the interaction between income redistribution and changes in income volatility. Recent papers (such as Auclert, 2019; Auclert et al., 2018; Debortoli and Galí, 2022; Wolf, 2021) have highlighted the role of sufficient statistics as a way to reduce the dimensionality of HA models and to summarize the equilibrium effects of fiscal and monetary policy. We contribute further by showing that the heteroskedasticity channel of output responses to demand shocks can be suitably captured by a few observable sufficient statistics.

There is a large literature on the theory of income risk and market incompleteness. In particular, within a CARA utility framework, Caballero (1990) shows how income risk produces a drift in the consumption process leading to excess consumption growth. Wang (2002) enriches the CARA model by Caballero (1990) to study the role of conditionally heteroskedastic income, postulating a linear relation between conditional income volatility and lagged earnings. Compared to those papers, our model, while making use of CARA utility, imposes only a differentiability requirement to the conditional variance (as a function of lagged earnings), without restricting the dependence to be linear. Wang (2003) and Acharya and Dogra (2020) combine the consumption model of Caballero (1990) with a general equilibrium setting. Crucially, our work differs from these papers by showing the general equilibrium effects of heteroskedasticity.

The HANK setup of Bayer et al. (2019) assumes a heteroskedastic process for labor productivity where the evolution of the variance over time depends on exogenous shocks. Unlike them, we focus on the cross-sectional dimension and we take heteroskedasticity as endogenously determined by

[7] Consistently with previous work, e.g., Broer et al. (2022), Guvenen et al. (2017), Parker et al. (2010).

the income distribution. Bilbiie (2018) builds a two-agent NK model augmented with idiosyncratic risk, modeled through a Markov transition structure between hand-to-mouth (HtM) and non-HtM states where the transition probabilities co-move with aggregate output. Since non-HtM agents are identical across them, they all face the same individual risk and aggregate fluctuations are linked to the cyclical property of the transition probabilities. In addition, Acharya and Dogra (2020) build a CARA utility model with a Gaussian income process where the volatility is identical across households, but depends over time on aggregate output, capturing the cyclicality of individual income risk. Unlike these papers, we assume that individual income is heteroskedastic in the cross-section and we show that cross-sectional heterogeneity in idiosyncratic risk interacts with the earnings distribution, thereby affecting aggregate fluctuations in response to demand shocks.

Debortoli and Galí (2022) study the effects of heterogeneous changes in individual consumption risk over the business cycle. Their setup has two main features. First, a precautionary motive due to incomplete insurance. Second, heterogeneity in MPCs. In their model, this last feature is due to heterogeneous assets positions, hence to a heterogeneous distance from a natural borrowing limit. Differently from them, in our model heterogeneity in MPCs is rooted in heteroskedasticity.

In addition, we contribute to the literature on *tractable* heterogeneous agent models (for instance, Acharya et al., 2020; Acharya and Dogra, 2020; Bilbiie, 2008, 2018, 2020; Bilbiie et al., 2013; Challe, 2020; Challe and Ragot, 2016; Ravn and Sterk, 2021) building a general equilibrium HA framework where we integrate heteroskedasticity and idiosyncratic risk with heterogeneity in MPCs and a non-degenerate wealth distribution in an analytically tractable and transparent way.

Recently, a large debate concerning the forward guidance puzzle (FGP), starting from Del Negro et al. (2015), focused on precautionary savings and market incompleteness as a way to solve the puzzle (for instance, see McKay et al., 2016, 2017). On the contrary, Werning (2015) argues that market incompleteness per se is not sufficient to solve the FGP and shows that risk counter-cyclicality may exacerbate the puzzle. Bilbiie (2018) summarizes these arguments, showing that pro-cyclical inequality and/or pro-cyclical risk can solve the FGP. We claim that the interaction between income redistribution (following positive news) and marginal changes in conditional income volatility can mitigate the FGP, although only if our covariance sufficient statistic is positive.

The rest of the paper is structured as follows. Section 2 describes the empirical evidence on cross-sectional heteroskedasticity. Section 3 presents the model. Section 4 provides estimates of key sufficient statistics characterizing the model. Section 5 concludes.

2 Evidence on heteroskedasticity

A large empirical evidence supports the hypothesis of cross-sectional heteroskedasticity in the individual income process.[8] Moreover, recent contributions in the literature on non-parametric esti-

[8] See Meghir and Pistaferri (2004).

mation of the income process show the existence of a relation between the volatility of the income process and the percentiles of the lagged income distribution.[9] Guvenen et al. (2021) find evidence of an asymmetric U-shaped relation between the standard deviation of the two-year log-earnings growth and the percentiles of the lagged earnings distribution.[10] In particular, the standard deviation is decreasing and convex up to the 95th percentile and sharply increasing for higher percentiles. Arellano et al. (2021) also study the moments of the income process, with Spanish administrative data, using an approach akin to the one in Guvenen et al. (2021). They find that the standard deviation of two-year log-earnings growth is decreasing over the percentiles of lagged income.

Consistent with this literature, we provide evidence of heterogeneity in the standard deviation of log-earnings across the earnings distribution. We use data from the waves of the Panel Survey of Income Dynamics (PSID) during the period 1969-2019 to estimate the standard deviation of the income process over lagged income bins.[11] We focus on labor earnings of family heads aged between 25 and 65, dropping low-earnings observations (annual earnings less than $1,000 in 2012 dollars). Our observations are at a biannual frequency.

To estimate income volatilities, for each year, we first split observations in deciles of lagged earnings.[12] Second, we filter log-earnings through a constant and a linear time trend, both specific to each lagged income decile.[13] Then, we compute (for each year in our sample) the standard deviation of filtered log-earnings conditional on the deciles of the lagged earnings distribution. Finally, we average the estimated standard deviations (for each lagged earnings bin) over time.

Figure 1 shows the estimated standard deviation of the income process, for each decile of the lagged earnings distribution. The observed pattern is similar to the one found in Guvenen et al. (2021) and based on administrative data.[14] The volatility of the income process is heterogeneous and depends on the position of a given individual over the earnings distribution. In particular, the standard deviation of log-earnings is decreasing up to the 8th decile of the earnings distribution and increasing over the last two deciles. Moreover, the decline in the standard deviation over the earnings distribution is large in magnitude: the standard deviation is halved as an individual moves from the 1st to the 8th decile.

[9] See Arellano (2014), Arellano et al. (2017), Arellano and Bonhomme (2017), and Guvenen et al. (2014).

[10] Guvenen et al. (2021) employ a large panel dataset drawing a representative 10% sample of the U.S. population from the Social Security Administration (SSA), and measure the presence of non-linearities and non-normalities in individual earnings dynamics over the life cycle. They document the existence of a significant empirical relation between higher moments of the income process (i.e., standard deviation, skewness, kurtosis) and age/percentiles of the income distribution.

[11] See Appendix B.1 for a more detailed description of the data.

[12] We define lagged earnings as earnings two years earlier.

[13] We also condition on age and gender.

[14] See Figure C.1 (left panel) in the Online Appendix in Guvenen et al. (2021).

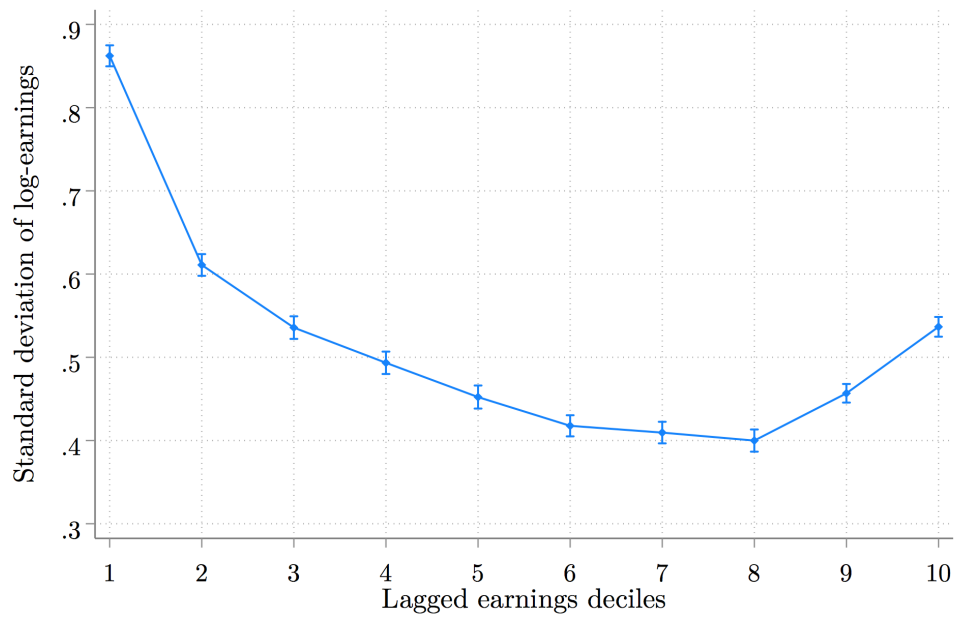

FIGURE 1: ESTIMATES OF THE STANDARD DEVIATION OF LOG-EARNINGS

Notes. The figure shows the standard deviation of filtered log-earnings, for each decile of the distribution of (two-years) lagged earnings. Filtered log-earnings are the residuals of a regression of log-earnings on a constant and a linear time trend, specific to each lagged earnings decile (controlling also for age and gender indicators). We report confidence intervals at the 90% level.

3 A heteroskedastic New Keynesian model

In this section, and motivated by the empirical evidence described in Figure 1, we build a general equilibrium model with heterogeneous agents and a heteroskedastic income process.

Consider a continuum of measure 1 of agents/households in the interval $[0,1]$ indexed by $i \in [0,1]$. Let $y_{i,t}$ denote the income of agent i and assume that the earnings process can be characterized as the sum of two components:

$$y_{i,t} = y^d_{i,t} + \epsilon_{i,t}, \tag{1}$$

where $y^d_{i,t}$ and $\epsilon_{i,t}$ denote the deterministic and the stochastic components of income respectively. We make the following assumptions.

Assumption 1. *For every agent $i \in [0,1]$, the path of the deterministic component of income $\left(y^d_{i,t}\right)_{t=0}^{\infty}$ is known in every period $t \geq 0$.*

Assumption 2. *For every agent $i \in [0,1]$, the process for the stochastic component is Gaussian, conditional on time $t-1$ information set $\mathcal{I}_{t-1}$, with its variance being a function of lagged deterministic income:*

$$\epsilon_{i,t} \mid \mathcal{I}_{t-1} \sim \mathcal{N}(0, \sigma^2_{i,t}), \quad \sigma^2_{i,t} := \sigma^2(y^d_{i,t-1}). \tag{2}$$

Assumption 3. *The random variables* $(\epsilon_{i,t})_{i\in[0,1]}$ *are conditionally stochastically independent:*

$$\epsilon_{i,t} \perp\!\!\!\perp \epsilon_{j,t} \mid \mathcal{I}_{t-1} \quad \forall i,j \in [0,1],\ i \neq j.$$

Assumption 3 is a standard requirement of cross-sectional independence for the stochastic components of the income process. Assumption 2 is the main requirement that we impose on the income process. Consistently with our empirical evidence of cross-sectional heteroskedasticity in income (see Figure 1), where the variance of the *unpredicatable* component varies over the income distribution, we assume that the unpredictable component $\epsilon_{i,t}$ of the income process is heteroskedastic.[15]

Note that the path of deterministic income being known by the agent (Assumption 1) and the conditional volatility of the stochastic component being a function solely of lagged deterministic earnings (Assumption 2) make the entire path of variances $\left(\sigma^2_{i,\tau}\right)_{\tau=t}^{\infty}$ known by the agent i at time t. In turn, these assumptions imply that the drift of the optimal consumption process is deterministic.

We argue that this assumption allows for large gains in terms of analytical tractability without loss of generality. Alternatively, we may in fact consider an identical two-period version of our infinite-horizon CARA-Gaussian model, and assume that the variance of the income process is a function of the *entire* lagged income (not only of its deterministic component, but also of the stochastic one). Suppose that individual income is stochastic and normally distributed with mean y_t and variance $\sigma^2_{i,t}$: $y_{i,t} \mid \mathcal{I}_{t-1} \sim \mathcal{N}(y_t, \sigma^2_{i,t})$. Then, one could model income volatility as: $\sigma^2_{i,t} := \sigma^2(y_{i,t-1})$. One can show that the two-period version of the model admits a closed-form solution even under this relaxed parametric assumption on the overall income process. Moreover, one can also prove that the implications of the infinite horizon and the two-period models are identical and that the main results of the paper can be derived (and coincide) in both frameworks.[16]

3.1 Households

In every period t, each household can purchase a risk-free (real) bond $a_{i,t+1}$ (paying one unit of good in $t+1$) at the price $1/(1+r_t)$. Additionally, the household can consume a perishable consumption good $c_{i,t}$, using its total (net) income $y_{i,t}$ and its beginning-of-period wealth $a_{i,t}$.

Households maximize their intertemporal utility – discounted with a factor $\beta \in (0,1)$ – over an infinite-horizon, choosing consumption and savings (labor supply is inelastic) under the sequence of budget constraints and the initial condition $a_{i,0}$ on bond holdings.[17] The problem can be formulated

[15] While Figure 1 looks at log-earnings, without loss of generality, we focus on earnings in *levels*, as it is natural in the CARA utility setting. See Caballero (1990), Wang (2002), Wang (2003), and Acharya and Dogra (2020).

[16] These results are available upon requests from the authors.

[17] The usual no-Ponzi-game condition applies as an additional constraint:

$$\lim_{T\to\infty} \prod_{t=0}^{T} \frac{1}{1+r_t} a_{i,T+1} = 0.$$

recursively with the Bellman operator:

$$V_t(a_{i,t}, y^d_{i,t}, \epsilon_{i,t}) := \max_{(c_{i,t}, a_{i,t+1})} \left\{ u(c_{i,t}) + \beta \cdot \mathbb{E}_t \left[V_{t+1}(a_{i,t+1}, y^d_{i,t+1}, \epsilon_{i,t+1}) \right] \right\}$$
$$\text{s.t.} \quad c_{i,t} + \frac{a_{i,t+1}}{1+r_t} = y_{i,t} + a_{i,t},$$
$$a_{i,0} \quad \text{given},$$

where $\mathbb{E}_t(\cdot) := \mathbb{E}(\cdot \mid \mathcal{I}_t)$ is the expectation operator conditional on time t information set and $V_t(\cdot)$ is the time t value function. In addition to the endogenous state $a_{i,t}$, the exogenous states $y^d_{i,t}$ and $\epsilon_{i,t}$ are relevant for the household's decision, since $y^d_{i,t}$ determines the volatility of $\epsilon_{i,t+1}$, while $\epsilon_{i,t}$ is the current realization of the income shock.

3.1.1 Solution to the consumer problem

Let $u(\cdot)$ be a CARA - exponential utility function: $u(c) = -\frac{1}{\gamma}\exp\{-\gamma c\}$ with γ being the coefficient of absolute risk aversion. The Euler equation derived from the utility maximization problem is:

$$\exp\{-\gamma c_{i,t}\} = \beta(1+r_t) \cdot \mathbb{E}_t\big[\exp\{-\gamma c_{i,t+1}\}\big]. \tag{3}$$

The solution of the problem is characterized by equation (3), together with the sequence of budget constraints, the initial condition and the transversality condition on bonds:

$$\lim_{T\to\infty} \mathbb{E}_t\big[\beta^T u'(c_{i,T}) \cdot a_{i,T}\big] = 0.$$

A procedure to find a closed-from solution for consumption is proposed by Caballero (1990) and is based on a "guess and verify" of the stochastic process followed by optimal consumption.[18] We extend that approach to the more general case in which r_t and the variance of the income process are not constant over time, and $\beta \neq 1 + r_t$.

Proposition 1. *The optimal consumption plan follows the stochastic difference equation:*

$$c_{i,t} = c_{i,t+1} - \frac{1}{\gamma}\log[\beta(1+r_t)] - \frac{\gamma\mu^2_{t+1}}{2}\sigma^2_{i,t+1} - \mu_{t+1}\epsilon_{i,t+1}, \tag{4}$$

where μ_t, the MPC out of wealth, follows the recursion:

$$\mu_t = \frac{\mu_{t+1}(1+r_t)}{1+\mu_{t+1}(1+r_t)}. \tag{5}$$

Proof. See Appendix A.1.

[18] An alternative solution method is followed by Acharya and Dogra (2020). They guess a solution for consumption of the form $c_{i,t} = \mathcal{C}_t + \mu_t(y_{i,t} + a_{i,t})$ and find the value of the coefficients $\mathcal{C}_t, \mu_t$ recursively.

Equation (4) is worth a few considerations. Notice that, relative to a standard permanent-income hypothesis (PIH) specification, the volatility term $(\gamma\mu_{t+1}^2/2)\cdot\sigma_{i,t+1}^2$ captures the effects of precautionary savings. Higher future income volatility depresses current consumption since agents have an incentive to save more as a form of partial insurance. In our setting, however, and unlike Acharya and Dogra (2020) who derive a similar equation in a homoskedastic CARA framework, the variance of future earnings that appears in (4) is *agent specific* since it is a function of current household income and it is, therefore, heterogeneous across agents.

3.2 Heterogeneity in MPCs

As in the corresponding homoskedastic CARA framework, MPCs in our model depend on current and future real interest rates. An increase in the real interest rate makes MPCs higher as a consequence of a stronger income effect (as can be seen from equation (5)). Consider a marginal increase in earnings: the household optimally saves part of it (or reduces its borrowing) obtaining higher future returns (since the interest rate has increased). These higher returns determine a positive income effect that optimally induces the agent to spend a larger fraction of the income shock (i.e., the MPC is higher).

In our heteroskedastic framework, however, there are two further implications for MPCs. First, there is a distinction between the MPC out of *wealth*, that coincides with μ_t (and with the MPC in the homoskedastic case) and the MPC out of a transitory *income* shock.[19] Second, there exists a non-degenerate distribution of MPCs out of transitory income, since different positions over the earnings distribution affect future income volatility and the precautionary savings motive. We summarize these findings in the following proposition.

Proposition 2. *Assume that $\sigma^2:\mathbb{R}\to\mathbb{R}_+$ is differentiable with respect to lagged (deterministic) income. Then:*

(i) MPCs out of wealth are identical across agents and differ from MPCs out of transitory income if the marginal change in future income volatility out of current earnings is non-null, i.e., if $\partial\sigma_{i,t+1}^2/\partial y_{i,t}\neq 0$.[20]

(ii) MPCs out of transitory income are heterogeneous across agents if the marginal change in future income volatility out of current earnings is non-constant over the earnings distribution, i.e., if $\partial\sigma_{i,t+1}^2/\partial y_{i,t}$ is non-constant.

In particular:

$$MPC_{i,t} := \frac{\partial c_{i,t}}{\partial y_{i,t}} = \mu_t - \frac{1}{1+\mu_{t+1}(1+r_t)}\frac{\gamma\mu_{t+1}^2}{2}\frac{\partial\sigma_{i,t+1}^2}{\partial y_{i,t}}. \tag{6}$$

[19] We define a transitory income shock as a "MIT shock" to the deterministic component of income.

[20] Notice that this derivative should be taken with respect to the deterministic component of the income process. To simplify notation and exposition of our results, we write $y_{i,t}$ instead of $y_{i,t}^d$ henceforth.

Proof. See Appendix A.1.

Interestingly, our framework is able to combine three different characteristics of the heterogeneous agents literature in a tractable way: (i) precautionary savings (captured by the variance term appearing in equation (4)), (ii) heterogeneity in MPCs out of transitory income, and (iii) difference between MPCs out of wealth and income.[21] These results contrast with the standard homoskedastic framework, where MPCs out of income are identical across agents and coincide with the MPC out of wealth.

Notice that our empirical evidence from Figure 1 suggests that $\partial\sigma^2_{i,t+1}/\partial y_{i,t}$ is both *negative* and *increasing* up to the 8th decile of the income distribution. Consistent with this fact, and assuming that income volatility $\sigma^2_{i,t+1}$ is a decreasing and convex function of lagged income, it is also immediate to reproduce the finding (of some empirical literature on MPCs) whereby MPCs are decreasing over the earnings distribution.

Lemma 1. *If the variance of individual income $\sigma^2 : \mathbb{R} \to \mathbb{R}_+$ is twice differentiable and strictly convex in lagged income, then the marginal propensity to consume of agent i is decreasing in income:*

$$\frac{\partial MPC_{i,t}}{\partial y_{i,t}} < 0. \tag{7}$$

The result from Lemma 1 is easy to obtain from equation (6). In particular:

$$\frac{\partial MPC_{i,t}}{\partial y_{i,t}} = -\frac{1}{1+\mu_{t+1}(1+r_t)}\frac{\gamma\mu^2_{t+1}}{2}\cdot\underbrace{\frac{\partial^2\sigma^2_{i,t+1}}{(\partial y_{i,t})^2}}_{>0} < 0.$$

Intuitively, if the variance of future income is *decreasing* and *convex* in current income, a marginal increase in current income has the effect, at the margin, of weakening the precautionary saving motive relatively more for poor than rich households. Thus agents at the bottom percentiles of the income distribution feature a higher MPC and, at the margin, increase consumption relatively more than agents at the top percentiles.

3.3 Aggregation and general equilibrium

We aggregate consumption integrating with respect to the Lebesgue measure. Let $c_t := \int_{[0,1]} c_{i,t}\, di$ and let us assume that the function $\sigma^2 : \mathbb{R} \to \mathbb{R}_+$ is measurable and integrable. Then, integrating the recursion for agents' optimal consumption (equation (4)), we obtain the aggregate consumption function:

$$c_t = c_{t+1} - \frac{1}{\gamma}\log[\beta(1+r_t)] - \frac{\gamma\mu^2_{t+1}}{2}\int_{[0,1]}\sigma^2_{i,t+1}di. \tag{8}$$

[21] MPCs out of housing wealth (see Guren et al., 2021) and out of financial wealth (see Chodorow-Reich et al., 2021) are estimated to be 2.67 cents and 3.2 cents per dollar, respectively, i.e., significantly smaller than usual estimates of MPCs out of income.

Note that the previous result can be obtained by a continuous version of the Law of Large Numbers (LLN)[22] since the integral of the income process innovation can be thought of as the limit of the means, converging to the population average:

$$\int_{[0,1]} \epsilon_{i,t+1} \, di = 0 = \mathbb{E}_t(\epsilon_{i,t+1}).$$

The economy has a supply-side that produces aggregate output y_t and a government that purchases part of this output, through government spending g_t.[23] Therefore, the goods market clearing requires $y_t = c_t + g_t$. Imposing this condition, we obtain a dynamic aggregate demand (AD) equation:

$$y_t = y_{t+1} - \frac{1}{\gamma} \log[\beta(1 + r_t)] - \frac{\gamma \mu_{t+1}^2}{2} \int_{[0,1]} \sigma_{i,t+1}^2 \, di + g_t - g_{t+1}. \tag{9}$$

In addition, the real and the nominal side of the economy are assumed to be generically linked via a Phillips-curve relation between current inflation π_t, future inflation, and output (implicitly, through the function $P(\cdot)$):

$$P(\pi_t, \pi_{t+1}, y_t) = 0. \tag{10}$$

Finally, the monetary authority follows a given (active) monetary policy rule. We consider two different specifications for monetary policy.

1. *Interest rate peg.* The monetary authority pegs the real interest rate to its steady-state value r:

$$r_t = r + v_t, \tag{11}$$

   where v_t is a stochastic process characterizing exogenous monetary shocks.

2. *Taylor rule.* The monetary authority sets the nominal interest rate i_t as function of current inflation, through the feedback Taylor coefficient $\phi_\pi > 0$:

$$1 + i_t = (1 + r) \cdot (1 + \pi_t)^{\phi_\pi} \exp(v_t). \tag{12}$$

3.3.1 Equilibrium

Given a process for $(g_t, v_t)_{t=0}^{\infty}$, an equilibrium is an allocation $(y_t)_{t=0}^{\infty}$, a vector of prices $(i_t, \pi_t, r_t)_{t=0}^{\infty}$ and a sequence of MPCs out of wealth $(\mu_t)_{t=0}^{\infty}$ such that:

- the dynamic AD – equation (9) – holds;
- the MPCs out of wealth recursion – equation (5) – holds;
- the PC – equation (10) – holds;

[22] See Uhlig (1988), Proposition 1.
[23] See Appendix A.2.

- monetary policy follows a given monetary policy rule, either the peg rule (equation (11)) or the Taylor rule (equation (12));
- the Fisher equation holds:

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}}.$$

3.4 Sufficient statistics

We are now ready to establish one of our key results. We show that the equilibrium response of output to alternative demand shocks can be fully described through a small set of observable sufficient statistics, capturing the heterogeneous consumption choices of different agents. Before turning to a more general characterization under price rigidity, we assume below that monetary policy keeps a constant real interest rate (up to an exogenous shock v_t).[24]

Proposition 3. *The equilibrium response of aggregate output to either government spending or real interest rate shocks is captured by the following cross-sectional sufficient statistics:*

(i) Expected marginal change in the income volatility out of individual earnings:

$$\underbrace{\mathbb{E}\left(\frac{\partial \sigma^2_{i,t+1}}{\partial y_{i,t}}\right)}_{\textit{individual risk effect}}.$$

(ii) Expected (individual) earnings sensitivity to aggregate output:

$$\underbrace{\mathbb{E}\left(\frac{\partial y_{i,t}}{\partial y_t}\right)}_{\textit{income cyclicality effect}}.$$

(iii) Covariance between the marginal change in (individual) income volatility out of (individual) earnings and the (individual) earnings sensitivity to aggregate output:

$$\underbrace{Cov\left(\frac{\partial \sigma^2_{i,t+1}}{\partial y_{i,t}}, \frac{\partial y_{i,t}}{\partial y_t}\right)}_{\textit{risk-cyclicality covariance effect}}.$$

Moreover, the output multiplier of a fully transitory government spending shock is:

$$\Theta_t := \frac{\partial y_t}{\partial g_t} = \left\{1 + \frac{\gamma \mu^2_{t+1}}{2}\left[\mathbb{E}\left(\frac{\partial \sigma^2_{i,t+1}}{\partial y_{i,t}}\right) \cdot \mathbb{E}\left(\frac{\partial y_{i,t}}{\partial y_t}\right) + Cov\left(\frac{\partial \sigma^2_{i,t+1}}{\partial y_{i,t}}, \frac{\partial y_{i,t}}{\partial y_t}\right)\right]\right\}^{-1}, \tag{13}$$

[24] Henceforth we also assume that it is possible to differentiate the variance function with respect to lagged income and to exchange the derivative and the integral.

while the output multiplier of a fully transitory real interest rate shock is:

$$\frac{\partial y_t}{\partial r_t} = -\Theta_t \cdot \frac{1}{\gamma(1+r_t)}. \tag{14}$$

Proof. See Appendix A.1.

Consider statistics (i)-(iii). All combined they capture how individual *risk* varies over the business cycle and how this effect is heterogeneous across the income distribution.

Statistic (i) captures the expected marginal effect of realized income on individual risk (*individual risk effect*). Statistic (ii) captures the degree of cyclicality of individual income with respect to aggregate income (*income cyclicality effect*). Finally, statistic (iii) describes how the effect on individual risk co-varies with the degree of income cyclicality (*risk-cyclicality covariance effect*). Hence, the size of both the government spending and real interest rate multipliers depends on the interaction between the cross-sectional variation in the precautionary saving motive and how the distribution of earnings is shaped by aggregate output.

Consider the (product of the) first two sufficient statistics in (13), i.e., (i) the individual risk effect and (ii) the income cyclicality effect. Suppose that a negative demand shock decreases output on impact and that, in turn, individual earnings decline. If, *on average*, individual risk is decreasing in individual income (so that the product of (i) and (ii) is negative),[25] following the negative shock, agents face *more risk* on average and therefore feature a larger precautionary saving motive. Therefore, they decrease consumption relatively more than if average risk were unaffected.

However, the latter precautionary savings response is *not homogeneous* across agents. This can be gauged from statistic (iii), which captures the risk-cyclicality covariance effect. The latter statistic describes the interaction between how individual risk varies at the margin and how sensitive earnings are to aggregate output over the income distribution. Intuitively, consider a negative demand shock. As aggregate income decreases, individual earnings may react in the cross-section in a heterogeneous fashion. Then, changes in individual earnings affect future income volatility, and in turn precautionary savings and individual MPCs. In turn, the heterogeneous consumption response at the margin feeds back into aggregate demand. By way of example, suppose that idiosyncratic risk is decreasing in individual income, i.e., $\partial\sigma^2_{i,t+1}/\partial y_{i,t} < 0$. If those agents whose income is more sensitive to aggregate income are also those for whom risk increases (at the margin) relatively more (following a decrease in their individual earnings), then the covariance term is negative, leading to amplification. In other words, amplification of a negative demand shock requires that those agents whose income is more sensitive to aggregate income are also those with a stronger increase in the precautionary saving motive and higher MPCs. Noticeably, our estimation in Section 4.3 suggests a negative value for the covariance statistic, providing supporting evidence for overall amplification of demand shocks.

[25] This is consistent with our empirical estimates described in Section 4 below.

3.4.1 Homoskedasticity: a special case

It is useful to compare our results so far to a standard setting with homoskedasticity (i.e., assuming that the volatility of the income process is common for all the households). Under (cross-sectional) homoskedasticity, let us denote the common volatility of the income process as σ_t^2. The dynamic AD equation then becomes:

$$y_t = y_{t+1} - \frac{1}{\gamma}\log[\beta(1+r_t)] - \frac{\gamma\mu_{t+1}^2}{2}\sigma_{t+1}^2 + g_t - g_{t+1}.$$

Since σ_{t+1}^2 is identical across the earnings distribution, changes in individual income (deriving from changes in output) neither affect risk, nor precautionary savings, nor MPCs at the margin. Therefore:

$$\mathbb{E}\left(\frac{\partial\sigma_{i,t+1}^2}{\partial y_{i,t}}\right) = Cov\left(\frac{\partial\sigma_{i,t+1}^2}{\partial y_{i,t}}, \frac{\partial y_{i,t}}{\partial y_t}\right) = 0,$$

and the output multiplier of an aggregate demand shock becomes, under homoskedasticity:

$$\Theta_t^{hom} := \frac{\partial y_t}{\partial g_t} = 1.$$

Hence, depending on the sign of the three sufficient statistics outlined above, the aggregate output multiplier of a temporary demand shock under heteroskedasticity, Θ_t, could either exceed or not the multiplier under homoskedasticity, $\Theta_t^{hom} = 1$. Notice also that in the homoskedastic case the output multiplier of an aggregate demand shock coincides with the one in a representative-agent economy.[26]

3.5 Results in the general model

Let us turn to a more general analysis of the first-order effects of fiscal (government spending) and monetary shocks. Let us suppose that monetary policy follows the Taylor rule specification of equation (12).

3.5.1 Steady state

We can approximate the model around a zero-inflation steady state. We set government spending in the steady state equal to zero and we normalize aggregate consumption and output to 1. Hence:

$$y = c = 1, \quad g = 0, \quad \pi = 0.$$

[26] That is equal to 1. See Woodford (2011).

Let $F : \mathbb{R} \to [0,1]$ be the cumulative density function describing the stationary distribution of (deterministic) income, y_i^d.[27] Under some additional assumptions, we can prove that:

$$\bar{\sigma}^2 := \int_{\mathbb{R}} \sigma^2(y)\ dF(y) = \int_{[0,1]} \sigma^2(y_i^d)\ di, \tag{15}$$

where $\bar{\sigma}^2$ is defined as the average income volatility in the steady state. Moreover, the steady-state value for the real interest rate is determined from the dynamic AD equation (9):

$$\left(\frac{1+r}{r}\right)^2 \log[\beta(1+r)] = -\frac{\gamma^2}{2}\bar{\sigma}^2. \tag{16}$$

Assuming that $\bar{\sigma}^2 > 0$,[28] a solution $r > 0$ to this equation exists and is unique.

Given the steady-state value for the real interest rate, the steady-state MPC out of wealth is determined from the MPC recursion, described in equation (5):

$$\mu = \frac{r}{1+r}. \tag{17}$$

Finally, let us define Γ as a function of the sufficient statistics (i)-(iii) in the steady state:

$$\Gamma := \int_{[0,1]} \left(\frac{\partial \sigma_i^2}{\partial y_i}\frac{\partial y_i}{\partial y}\right) di = \mathbb{E}\left(\frac{\partial \sigma_i^2}{\partial y_i}\right) \cdot \mathbb{E}\left(\frac{\partial y_i}{\partial y}\right) + Cov\left(\frac{\partial \sigma_i^2}{\partial y_i}, \frac{\partial y_i}{\partial y}\right). \tag{18}$$

Notice that $\Gamma = 0$ holds in the two limit cases of a representative-agent and in the homoskedastic version of our model. Details and proofs related to the steady state are in Appendix A.3.

3.5.2 Log-linear representation

We take a log-linear approximation around the previously defined steady state. A *hat* notation denotes variables in deviation from their steady-state values: $\hat{x}_t := \frac{x_t - x}{x}$ and $\hat{g}_t := g_t$. Additionally, suppose that the Phillips Curve admits a standard forward-looking linear approximation (with k being its slope).[29] The linearized model can be described by the following four equations.[30]

$$\hat{y}_t = \Theta \cdot \left[\hat{y}_{t+1} - \frac{1}{\gamma} \cdot (\hat{i}_t - \hat{\pi}_{t+1}) - \Lambda \cdot \hat{\mu}_{t+1} + (\hat{g}_t - \hat{g}_{t+1})\right], \tag{19a}$$

$$\hat{\mu}_t = \tilde{\beta} \cdot [\hat{\mu}_{t+1} + (\hat{i}_t - \hat{\pi}_{t+1})], \tag{19b}$$

$$\hat{\pi}_t = \tilde{\beta} \cdot \hat{\pi}_{t+1} + k \cdot \hat{y}_t, \tag{19c}$$

$$\hat{i}_t = \phi_\pi \cdot \hat{\pi}_t + v_t, \tag{19d}$$

[27] We can always assume that such a function exists in the steady state.

[28] Equivalently, assuming that a set of agents with non-zero measure has strictly positive variances in the steady state.

[29] We linearize the PC obtained under convex price adjustment costs as in Rotemberg (1982). See Appendix A.2.4.

[30] Note that the log-linear Fisher equation $\hat{r}_t = \hat{i}_t - \hat{\pi}_{t+1}$ has already been replaced into the other equilibrium equations.

where:

$$\begin{aligned}
\Theta &:= \left(1 + \frac{\gamma\mu^2}{2}\Gamma\right)^{-1} > 0, \\
\Lambda &:= \gamma \cdot \mu^2 \cdot \bar{\sigma}^2 > 0, \\
\tilde{\beta} &:= \frac{1}{1+r}.
\end{aligned} \tag{20}$$

Equation (19a) is the linearized version of the dynamic AD equation, equation (19b) is the linearized MPC recursion, equation (19c) is the linearized PC, and equation (19d) is the linearized Taylor rule. To conclude the description of the equilibrium, we assume that government spending and monetary shocks follow an AR(1) process:

$$\begin{aligned}
\hat{g}_t &= \rho_g \cdot \hat{g}_{t-1} + \epsilon_t^g, \\
v_t &= \rho_v \cdot v_{t-1} + \epsilon_t^v,
\end{aligned}$$

with $|\rho_g| < 1$, $|\rho_v| < 1$ and ϵ_t^g, ϵ_t^v being white noise serially uncorrelated processes, orthogonal between them, i.e., $\epsilon_t^g \perp \epsilon_t^v$.

Notice that parameter Λ in equation (19a) captures the (direct) marginal effect on output of changes in the MPC out of wealth. As in the CARA-homoskedastic model of Acharya and Dogra (2020), monetary policy is able to affect MPCs through interest rates. In addition, MPCs affect the pass-through of idiosyncratic income risk to consumption risk. Hence, a higher (future) real interest rate increases the (future) MPC and reduces output since income risk translates into a greater consumption risk (a larger fraction of income is spent and consumption inherits "more" of the volatility of income).

Heteroskedasticity and discounting/compounding

A key feature that distinguishes our framework is the presence, in the AD equation (19a), of the parameter Θ, given by the inverse of a linear transformation of Γ, in turn a function of the sufficient statistics outlined in Proposition 3. Θ also represents the steady-state output multiplier of government spending of Proposition 3. Notice that Θ augments the standard representative-agent AD equation with a *discount/compound* factor for future expectations (about $\hat{y}_{t+1}, \hat{g}_{t+1}$) and current shocks ($\hat{r}_t, \hat{g}_t$). Parameter Θ is greater (less) than $\Theta_t^{hom} = 1$ if the average marginal change in risk and/or the covariance statistics are negative (positive).[31] We explore below the implications of this compounding/discounting factor for two main issues: equilibrium determinacy and the size of output multipliers.

[31] Recall that the income cyclicality effect captured by statistic (ii) is always positive.

3.5.3 Equilibrium determinacy

The presence of the parameter Θ in (19a) affects the conditions for equilibrium determinacy, since it imposes restrictions on the parameter subspace where the model features a unique non-explosive local equilibrium solution. In general we can show that with larger values of Θ the conditions for equilibrium determinacy become more stringent. The reason is that expectations about future output get compounded more when Θ is high and equilibrium solutions tend to become explosive. Below we treat the issue of determinacy separately for the two specifications of monetary policy: an interest rate peg and a Taylor rule.

Determinacy with an interest rate peg

Let us consider the simple case of a monetary authority that follows an interest rate peg, i.e., $\hat{r}_t = 0$, and let us assume that prices are fully fixed, i.e., $\hat{\pi}_t = 0$ for every $t \geq 0$. Then, the equilibrium is described uniquely by the AD equation (19a):

$$\hat{y}_t = \Theta \cdot \left[\hat{y}_{t+1} + (\hat{g}_t - \hat{g}_{t+1})\right]. \tag{21}$$

Equilibrium determinacy requires AD-discounting: $\Theta < 1$. In fact, when expectations about future output are discounted, the effect of future fiscal shocks or future deviations of output from its steady-state level have a smaller impact on current output the further they occur in the future. Consequently, when iterating forward to solve for current output:

$$\hat{y}_t = \lim_{T\to\infty} \Theta^T \cdot \hat{y}_{t+T} + \sum_{k=1}^{\infty} \Theta^k \cdot (\hat{g}_{t+k-1} - \hat{g}_{t+k}).$$

If $\Theta < 1$, then the limit term is null and the equilibrium path of output $(\hat{y}_t)_{t\geq 0}$ is bounded.[32] On the contrary, if $\Theta > 1$, then the limit term diverges and the solution for current output is explosive.

Determinacy under a Taylor rule

Turning to the Taylor rule specification for monetary policy, determinacy requires stronger restrictions to the parameter space when Θ is larger. Again, the greater Θ, the larger the compounding effect of future expectations (or the less they are discounted) back to current output and the stronger the reaction of monetary policy must be to stabilize the economy. In particular, for equilibrium determinacy, the inflation feedback coefficient ϕ_π has to rise with higher values of Θ, meaning that the monetary authority needs to overreact to inflation with a larger increase in the nominal interest rate.

[32] We also need to postulate that the sequence of changes in government spending is bounded. Let $\Delta\hat{g}_{t+1} := \hat{g}_{t+1} - \hat{g}_t$, then, we assume that:

$$\sup_{k\geq 1} |\Delta\hat{g}_{t+k}| < \infty.$$

We reduce the dimensionality of the model from four to three equations by substituting (19d) in the other three equations of the linear system (19) and rewrite it in matrix form as below:

$$\begin{pmatrix} \hat{y}_{t+1} \\ \hat{\mu}_{t+1} \\ \hat{\pi}_{t+1} \end{pmatrix} = \underbrace{\begin{bmatrix} \Theta^{-1} + \frac{k}{\tilde{\beta}}(\gamma^{-1} - \Lambda) & \frac{\Lambda}{\tilde{\beta}} & (\gamma^{-1} - \Lambda)\left(\phi_\pi - \frac{1}{\tilde{\beta}}\right) \\ -\frac{k}{\tilde{\beta}} & \frac{1}{\tilde{\beta}} & \frac{1}{\tilde{\beta}} - \phi_\pi \\ -\frac{k}{\tilde{\beta}} & 0 & \frac{1}{\tilde{\beta}} \end{bmatrix}}_{\Omega:=} \begin{pmatrix} \hat{y}_t \\ \hat{\mu}_t \\ \hat{\pi}_t \end{pmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Delta\hat{g}_{t+1} + \begin{bmatrix} \gamma^{-1} - \Lambda \\ -1 \\ 0 \end{bmatrix} v_t, \tag{22}$$

where $\Delta\hat{g}_{t+1} = \hat{g}_{t+1} - \hat{g}_t$. With this specification there are three forward-looking variables, $\hat{y}_t, \hat{\mu}_t, \hat{\pi}_t$, hence determinacy requires the matrix Ω to have three eigenvalues outside the unit circle. See Appendix A.4 for details.

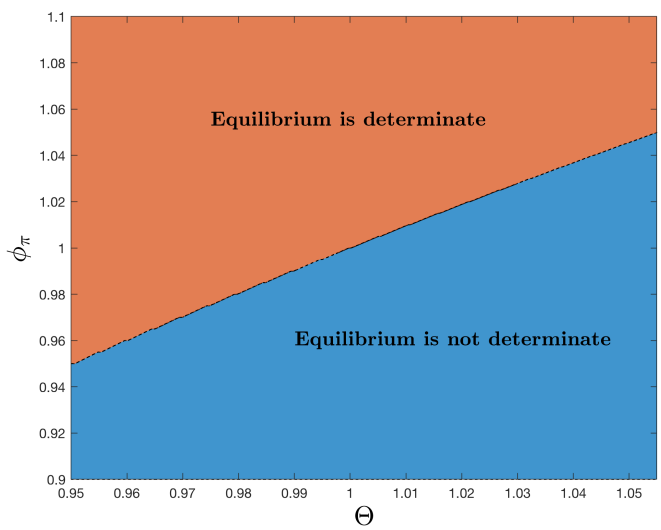

FIGURE 2: DETERMINACY REGION: Θ - ϕ_π.

Figure 2 plots the Taylor-rule coefficient ϕ_π as a function of Θ.[33] The area depicted in red corresponds to the region of the parameter space where the equilibrium is determinate. Notice that in the case $\Theta = 1$, corresponding to either the representative-agent or the homoskedasticity cases, the Taylor principle, i.e., $\phi_\pi > 1$, is both sufficient and necessary for determinacy. More generally, let $\bar{\phi}_\pi$ be the threshold for ϕ_π such that for every $\phi_\pi > \bar{\phi}_\pi$ the model has a determinate equilibrium. Then, $\bar{\phi}_\pi$ is increasing in Θ.

Summarizing, the presence in the AD equation (19a) of either compounding or discounting (through the parameter Θ) modifies the extent to which the Taylor principle ($\phi_\pi > 1$) is sufficient for local determinacy. In particular *compounding* – $\Theta > 1$ – requires monetary policy to react more strongly to an increase in output ($\bar{\phi}_\pi$ has to be greater than 1), while *discounting* – $\Theta < 1$ – allows

[33] To plot this figure and figures in Section 3.6 (i.e., Figures 3 and 4), we set the following values for the model's parameters: coefficient of absolute risk aversion $\gamma = -u''(c)/u'(c) = 3$; slope of the Phillips Curve $k = 0.1$; steady-state average income volatility $\bar{\sigma}^2 = 0.5$ (following Storesletten et al., 2004); discount factor $\beta = 0.96$.

for a weaker reaction of monetary policy (ϕ_π can be less than 1).

3.5.4 Analytical solution for multipliers

The model is tractable enough that we can easily derive analytical solutions. We solve for the impulse response functions (IRFs) to fiscal and monetary shocks through the method of undetermined coefficients. Let us suppose that the solution for output, inflation and MPCs follows the guess:

$$\hat{x}_t = \psi_x^u \cdot u_t \quad \text{where} \quad u \in \{g, v\}, \quad x \in \{y, \mu, \pi\}.$$

Substituting the guess in the system of equations (19) (substituting also for the Taylor rule), rearranging, and matching coefficients, we obtain the solution for the reduced form elasticities to both shocks.[34] The general expressions for the multipliers out of persistent shocks can be found in Appendix A.5.

All reduced-form multipliers become analytically transparent when fiscal and monetary shocks are white noise, i.e., $\rho_g = \rho_v = 0$. In this case, we have, for the *fiscal shock*:

$$\begin{aligned} \psi_y^g &= \frac{\gamma\Theta}{\gamma + \Theta \cdot \phi_\pi \cdot k} > 0, \\ \psi_\mu^g &= \phi_\pi \cdot \tilde{\beta} k \cdot \psi_y^g > 0, \\ \psi_\pi^g &= k \cdot \psi_y^g > 0, \end{aligned} \tag{23}$$

and for the *monetary shock*:

$$\begin{aligned} \psi_y^v &= -\frac{\Theta}{\gamma + \Theta \cdot \phi_\pi \cdot k} < 0, \\ \psi_\mu^v &= \tilde{\beta} + \phi_\pi \cdot \tilde{\beta} k \cdot \psi_y^v > 0, \\ \psi_\pi^v &= k \cdot \psi_y^v < 0. \end{aligned} \tag{24}$$

A positive shock to government spending increases output and inflation. Monetary policy responds by rising the nominal, and therefore the real interest rate, leading to a rise in MPCs. Higher MPCs have an offsetting contractionary effect on output since they amplify the pass-through of income to consumption risk. As for the monetary policy multiplier, a contractionary increase in the nominal interest rate reduces output and inflation, and causes the MPC to increase. Most importantly, the coefficients for output and inflation of fiscal and monetary shocks ($\psi_y^g, \psi_\pi^g, \psi_y^v, \psi_\pi^v$) are *increasing* in Θ in absolute value, consistent with the fact that higher values of Θ determine a general equilibrium amplification of demand shocks.

Note, again, that by imposing $\Theta = 1$ one obtains the IRFs for the model with homoskedasticity. It is therefore clear that $\Theta > 1$ generates amplification, while $\Theta < 1$ generates dampening of both

[34] We assume that $\phi_\pi > \rho_v, \rho_g$ and that $\gamma \cdot \Lambda < 1$. Both assumptions hold for our and, more generally, for empirically reasonable calibrations.

shocks:

$$\text{fiscal shock:} \quad |\psi_y^g(\Theta > 1)| > \underbrace{\frac{\gamma}{\gamma + \cdot\phi_\pi \cdot k}}_{\text{homosked.}} > |\psi_y^g(\Theta < 1)|,$$

$$\text{monetary shock:} \quad |\psi_y^v(\Theta > 1)| > \underbrace{\frac{1}{\gamma + \cdot\phi_\pi \cdot k}}_{\text{homosked.}} > |\psi_y^v(\Theta < 1)|.$$

3.6 Forward guidance puzzle

The expression *forward guidance puzzle* (FGP) (first used by Del Negro et al., 2015) denotes the puzzling prediction of standard New Keynesian models whereby changes in the nominal interest rate are more effective (on current output) the further they occur in the future. In this vein, news about monetary easing years ahead are at least as effective to boost economic activity as a cut in the current short-term interest rate. This puzzle occurs since agents perfectly anticipate the change in monetary policy and react by perfectly smoothing consumption over time. In addition, the positive feedback effect of monetary easing on output and inflation further reduces the real interest rate, amplifying the effects on current output of the announcement.

In our model, the FGP puzzle can be either worsened or solved, depending on the values of the sufficient statistics, that in turn determine the value of the parameters Γ and Θ (in the log-linear version, system of equations (19)).[35] Let us assume, for simplicity, that prices are fully fixed, i.e., $\hat{\pi}_t = 0$ for every $t \geq 0$ and consider a (credible) policy announcement of a nominal interest rate cut H periods ahead: $\hat{r}_{t+H} = \hat{i}_{t+H} = -\epsilon < 0$. We can rewrite the AD and the MPC recursions (equations (19a) and (19b)), iterating forward for H periods.

$$\hat{y}_t = \Theta^{H+1} \cdot \hat{y}_{t+H+1} - \frac{\Theta^{H+1}}{\gamma} \cdot \hat{i}_{t+H} - \Lambda \cdot \sum_{k=1}^{H+1} \Theta^k \hat{\mu}_{t+k}, \tag{25a}$$

$$\hat{\mu}_{t+k} = \tilde{\beta}^{H+1-k} \cdot \hat{\mu}_{t+H+1} + \tilde{\beta}^{H+1-k} \hat{i}_{t+H} \quad \text{for} \quad k = 0, ..., H. \tag{25b}$$

Additionally, using the same iterations, it is easy to show that $\hat{y}_{t+H+1} = \hat{\mu}_{t+H+1} = 0$. Substituting (25b) in (25a), we get:

$$\hat{y}_t = -\left[\frac{\Theta^{H+1}}{\gamma} + \Lambda \cdot \tilde{\beta}^{H+1} \sum_{k=1}^{H+1} \left(\frac{\Theta}{\tilde{\beta}}\right)^k\right] \cdot \hat{i}_{t+H} = \underbrace{\left[\frac{\Theta^{H+1}}{\gamma} + \Lambda \cdot \Theta^{H+1} \sum_{s=0}^{H} \left(\frac{\tilde{\beta}}{\Theta}\right)^s\right]}_{\Phi_H :=} \cdot \epsilon. \tag{26}$$

Let

$$\Phi_H := \left[\frac{\Theta^{H+1}}{\gamma} + \Lambda \cdot \Theta^{H+1} \sum_{s=0}^{H} \left(\frac{\tilde{\beta}}{\Theta}\right)^s\right] \tag{27}$$

[35] Recall that $\Gamma > 0 \iff \Theta < 1$.

be the *forward guidance (FG) multiplier*, expressed as a function of the time horizon H in which the interest rate cut will take place. Then, we can state that the FGP is "solved" when Φ_H is decreasing in H, in the sense that the further in the future the monetary easing is expected to occur, the less effective it will be on current output.

The formula for the FG multiplier can be decomposed in two parts, respectively describing two different channels through which future shocks affect current output.

- *Direct compounding/dampening* (direct C/D henceforth) channel, captured by the first term in expression (27):

$$\frac{\Theta^{H+1}}{\gamma}.$$

  This component describes the effects of future shocks on current output, *for a given path of MPCs*.

- *Change in MPCs* channel, captured by the second term in expression (27):

$$\Lambda \cdot \Theta^{H+1} \sum_{s=0}^{H} \left(\frac{\tilde{\beta}}{\Theta} \right)^{s}.$$

  This component describes the effects of future shocks on current output *through variations in the path of MPCs* (recall that in our model MPCs depend on current and future interest rates).

To study under which subset of the parameter space the FGP is either exacerbated or attenuated, we can take the derivative of Φ_H with respect to H and study its sign:[36]

$$\frac{\partial \Phi_H}{\partial H} = \Bigg[\underbrace{\log(\Theta) \cdot \frac{\Theta^{H+1}}{\gamma}}_{\text{derivative of 1}^{\text{st}}\text{ term in (27)}} + \underbrace{\frac{\Lambda\Theta}{\Theta - \tilde{\beta}} \left[\log(\Theta) \cdot \Theta^{H+1} - \log(\tilde{\beta}) \cdot \tilde{\beta}^{H+1} \right]}_{\text{derivative of 2}^{\text{nd}}\text{ term in (27)}} \Bigg].$$

In Figure 3 we show the contribution of each of the two channels (dashed and dotted lines for the direct C/D and the change in MPCs channels, respectively) and their cumulative effect on the FG multiplier (solid line) for different horizons H of the interest rate shock. We distinguish three scenarios for three different values of Θ (equivalently, of Γ, i.e., of our sufficient statistics). In the top panel we present the *discounting* case ($\Theta < 1$), in the middle one the homoskedastic benchmark ($\Theta = 1$) and in the bottom one the *compounding* scenario ($\Theta > 1$).

Notice that if $\Theta \geq 1$, then the second term in Φ_H is always increasing in H, while the first term is constant and increasing in H for $\Theta = 1$ and $\Theta > 1$, respectively. Therefore $\Theta \geq 1$ is a sufficient condition to exacerbate the FGP, as it can be seen from the bottom and middle panels in Figure 3.

[36] The summation in Φ_H can be rewritten as:

$$\Lambda \cdot \Theta^{H+1} \sum_{s=0}^{H} \left(\frac{\tilde{\beta}}{\Theta} \right)^{s} = \Lambda \cdot \Theta^{H+1} \frac{1 - \frac{\tilde{\beta}^{H+1}}{\Theta^{H+1}}}{1 - \frac{\tilde{\beta}}{\Theta}} = \frac{\Lambda\Theta}{\Theta - \tilde{\beta}} \left[\Theta^{H+1} - \tilde{\beta}^{H+1} \right].$$

In addition, notice that when the derivative of Φ_H with respect to H is positive, the FGP is exacerbated.

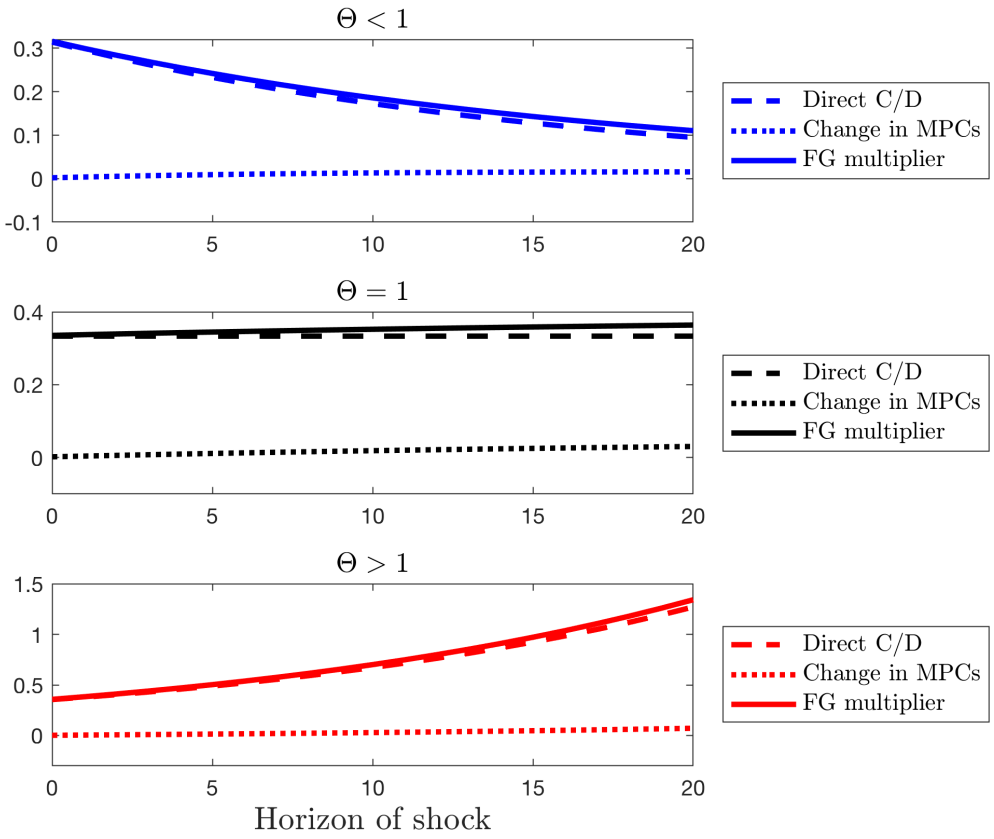

FIGURE 3: FG MULTIPLIERS: DISENTANGLING DIFFERENT CHANNELS.

Notes: The figure plots the "direct compounding/discounting (C/D)" channel, the "change in MPCs" channel, and the FG multiplier Φ_H as functions of the horizon H of the interest rate shock (all for three alternative values of Θ).

Intuitively, the second term in Φ_H captures the impact of news on current output through changes in the path of MPCs. In fact, as described by Acharya and Dogra (2020), news about future interest rate cuts reduce MPCs. Hence, since they weaken the pass-through from income to consumption risk, they have the effect of stimulating current demand. In addition, in every period, if $\Theta > 1$ the effect of a decline in MPCs is further amplified (through the demand amplification channel coming from heteroskedasticity described in Section 3.5.4) [37] and, then, compounded back to current output.

In addition to the previous MPC channel, suppose that $\Theta > 1$ (because, e.g., the correlation between the marginal change in risk and income sensitivities is negative). In this case, even keeping the path of MPCs fixed (i.e., isolating the effects of the "direct C/D channel"), positive news about an interest rate cut H periods ahead cumulate over time for two reasons. First, agents expect that the cut in $t + H$ will have an amplified effect on output in $t + H$ (through the usual demand

[37] This last amplification effect is absent in Acharya and Dogra (2020), while in our AD equation (19a) Λ is premultiplied by Θ.

amplification channel). Second, they also expect that output in $t + H - 1$ will increase more than one-to-one with y_{t+H} since one-period ahead positive news are also amplified (through the usual demand amplification channel). Moving backward period-by-period, the sequential amplification effect compounds back to current output. Therefore, the further the horizon H, the more compounding takes place, making current output more responsive to the forward guidance than to a current interest rate cut.

Conversely, when considering the case of $\Theta < 1$ the "direct C/D channel" is active in the opposite direction (see the dashed line of the top panel in Figure 3), in the sense that dampening takes place and future news are discounted over time. Hence, if the path of MPCs were kept constant, positive values for our covariance statistic[38] (equivalently, for $\Theta < 1$) would be able to solve the FGP.

Turning to the "change in MPC" channel when $\Theta < 1$: the second term in Φ_H has a non-linear shape in H, increasing for smaller values of H and decreasing thereafter.[39] Intuitively, for longer horizons, the dampening effect on output of changes in MPCs (since Θ premultiplies Λ in our AD equation) prevails on the expansionary consequences of declines in MPCs, while this prediction is reversed when the cut occurs over shorter horizons.

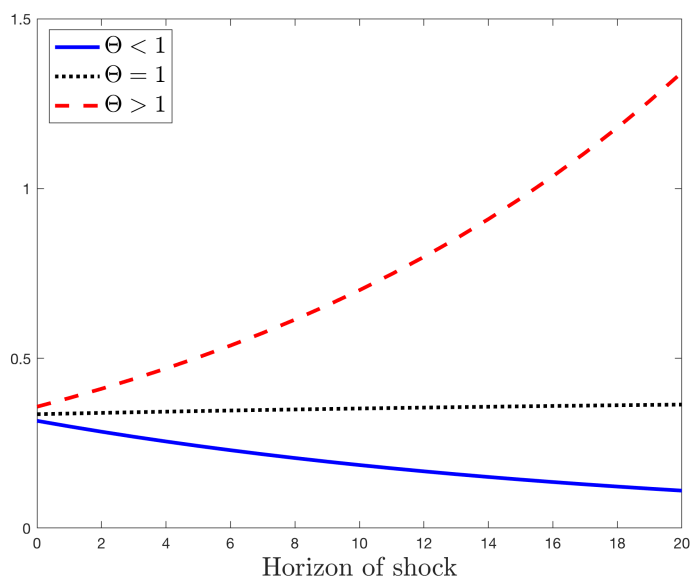

FIGURE 4: FG MULTIPLIERS

Notes: The figure plots the FG multiplier, Φ_H, as a function of the horizon H of the interest rate shock, for three alternative values of Θ.

[38] Also, positive values for the cross-sectional average change in individual income risk.

[39] Studying the sign of the second term in the derivative of the FPG multiplier out of the time horizon, with $\Theta < 1$ (and $\Theta \neq \tilde{\beta}$) we can find the threshold $\hat{H}$ at which its sign is reversed:

$$\left(\frac{\Theta}{\tilde{\beta}}\right)^{\hat{H}+1} = \frac{\log(\tilde{\beta})}{\log(\Theta)}.$$

For $H < \hat{H}$, the second term is increasing in H, while it decreases for $H > \hat{H}$.

Figure 4 summarizes the above-mentioned effects showing the FG multiplier for the cases of discounting (blue line) and compounding (red line) relative to the homoskedastic benchmark (black line). While $\Theta \geq 1$ is a sufficient condition to exacerbate the puzzle, $\Theta < 1$ may indeed solve the FGP. Notice, finally, that our model is not able to *simultaneously* attenuate the FGP and to generate output amplification of demand shocks.[40] In fact, attenuation of the FGP requires $\Theta < 1$, while demand amplification needs $\Theta > 1$. This incompatibility occurs since the underlying mechanism that regulates output responses to both future news and demand shocks is similar and depends on the interaction between marginal changes in income risk and changes in the earnings distribution over the business cycle.

4 Estimation

In this section we provide empirical evidence on the sufficient statistics that are central to Proposition 3. We use PSID biannual data from 1969 to 2019, as described in Section 2 and in Appendix B.1. In particular, we focus on labor earnings of family heads aged between 25 and 65, dropping low-earnings observations (annual earnings less than \$1,000 in 2012 dollars). As natural in the empirical literature on the income process, we work with variables in logs (e.g., log-earnings).[41]

We proceed in two steps. We first focus on estimating individual earnings sensitivities to fluctuations in aggregate income. Then we turn to the estimation of the sign of the individual risk effects across the income distribution.

4.1 Individual earnings sensitivities

We begin by estimating the degree of *cyclicality* of individual income, i.e., the elasticity of individual earnings to aggregate output. Standard estimation strategies followed in the literature regress log-earnings on log-GDP with coefficients that are specific to the relevant groups to which the individual belongs.[42] In particular, Guvenen et al. (2017) estimate individual income cyclicalities over the earnings distribution. Using administrative data from the SSA, they estimate a model in first-differences where log-earnings growth is regressed on log-GDP growth. Both the GDP coefficient and the intercept are specific to each lagged income bin. They find that income cyclicality follows a U-shaped pattern (decreasing up to the 90th percentile of the income distribution, increasing thereafter). Similarly, using CPS data, Parker et al. (2010) focus on the income cyclicality of individuals in the top 1 percent of the income distribution. More recently, Broer et al. (2022) find with German administrative data that low-income individuals have more cyclical earnings.[43] An alternative specification is followed by Schulhofer-Wohl (2011), which focuses on the income

[40] As common in the FGP literature with rational expectations, see Bilbiie (2018).

[41] We therefore re-define the sufficient statistics according to this log-specification.

[42] In our model, these groups are defined by lagged earnings bins.

[43] They find that this stronger cyclicality is mainly driven by the extensive margin (more countercyclical job loss for low earnings).

cyclicality of individuals with different degrees of risk-aversion.[44]

We specify our baseline empirical model in log-levels to remain closer to the structure of our theoretical model of Section 3. We then regress (individual) log real earnings – $\log(Y_{it})$ – on a constant, a linear time trend, log real GDP (linearly de-trended), age and gender controls (captured by the vector $Z_{i,t}$):

$$\log(Y_{i,t}) = \mu_g + \delta_g \cdot t + \beta_g \cdot \log(GDP_t) + Z'_{i,t} \cdot \zeta + \epsilon_{i,t}, \tag{28}$$

where the constant (μ_g) and the coefficients of the time trend (δ_g) and log-GDP (β_g) are specific to the lagged earnings bin, denoted by the subscript g, to which individual i belongs in a given year t. The coefficient β_g is the estimate of the average individual earnings sensitivity within lagged group g. Residuals are denoted by $\epsilon_{i,t}$. The model is estimated with pooled OLS (POLS).[45]

The blue dashed line (left axis) of Figure 5 plots the estimates of β_g across (two-year) lagged earnings quintiles.[46] Consistent with Guvenen et al. (2017), there is a clear decreasing pattern of earnings sensitivities over the lagged earnings distribution. Individuals belonging to lower quintiles of (lagged) earnings display a higher sensitivity of their individual income to aggregate income.

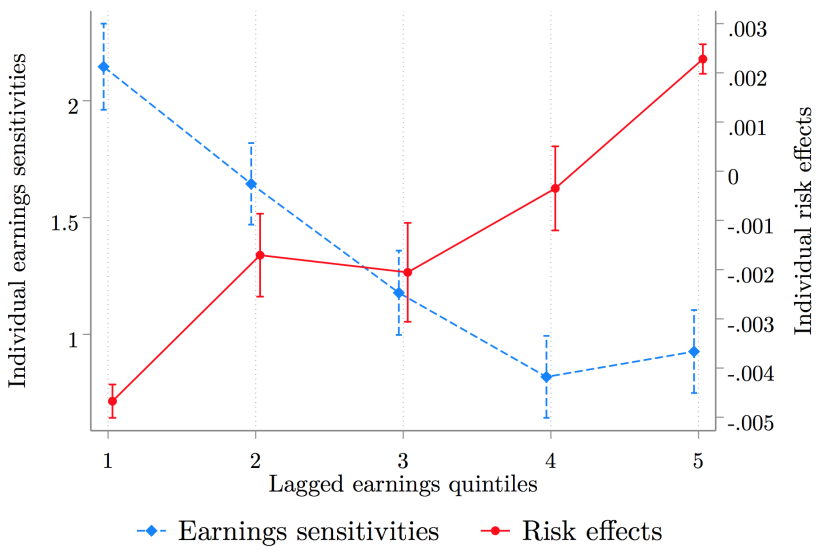

FIGURE 5: INDIVIDUAL EARNINGS SENSITIVITIES AND INDIVIDUAL RISK EFFECTS

Notes. The figure shows the cyclicality of the individual earnings β_g (left axis), as estimated from (28), and the estimate of the individual risk effect $\psi_g/100$ (right axis) for each quintile of the distribution of (two-years) lagged earnings. Confidence intervals at the 90% level are also shown.

[44] Schulhofer-Wohl (2011) estimates a random effect model with log-earnings in levels and log-PCE (real, per capita aggregate personal consumption expenditure), with PCE coefficients that are specific to different risk-aversion groups.

[45] The results are robust to a specification of the model in first-differences, to the inclusion of a fixed-effect estimator, and to a Bayesian bootstrap estimation procedure (see Appendix B.3).

[46] See also Table B-1 (column 1).

4.2 Individual risk effects

We define the *individual risk effect* as the marginal change in the *volatility* of log-earnings out of a percent change in lagged earnings. We estimate the individual risk effect through a two-step procedure: first, by estimating the conditional volatility of the earnings process for each lagged earnings bin g; second, by computing the marginal change of the volatility over the earnings distribution.

Conditional earnings volatilities. The conditional earnings volatility is defined as the variance of the unpredictable component of log-earnings, conditional on each lagged earnings bin g. To obtain conditional earnings volatilities we first estimate with POLS the model of equation (28), using two-years lagged earnings split into *deciles* (denoted as g^d), and we store the residuals ($\epsilon_{i,t}$).[47] Second, for each year t, we compute the sample variance of the residuals – denoted by $Var\left(\epsilon_{i,t}\,|g^d_{t-1}\right)$ – conditional on each lagged earnings bin g^d_{t-1}. In other words, for every year, we obtain ten estimates of the sample variance of residuals, one for each decile of the lagged earnings distribution.

Finally, for each lagged earnings bin g^d, we compute the arithmetic mean of these estimates over time and we denote it as $Var\left(\epsilon_{i,\cdot}\,|g^d\right)$.[48] As already shown in Figure 1, and consistent with the evidence in Guvenen et al. (2021) and Arellano et al. (2021), the volatility of the unpredictable component of log-earnings is decreasing over the (lagged) earnings distribution.

Individual risk effects. To estimate the individual risk effects we proceed in two steps. First – for each year t – we approximate the marginal change in the conditional earnings volatility ($\psi_{g,t}$) by computing the average change across two consecutive deciles, normalized by average earnings in the same two bins:

$$\begin{aligned}\psi_{g,t} &= \frac{\partial\sigma^2_{i,t}}{\partial\log(Y_{i,t-1})} = \frac{\partial\sigma^2_{i,t}}{\partial Y_{i,t-1}}\cdot Y_{i,t-1} \approx \\ &\approx \underbrace{\frac{Var\left(\epsilon_{i,t}\,|g^d_{t-1}\right) - Var\left(\epsilon_{i,t}\,|g^d_{t-1}-1\right)}{\bar{Y}(g^d_{t-1}) - \bar{Y}(g^d_{t-1}-1)}}_{\substack{\text{Average change in variance}\\ \text{for a unit of income}}} \cdot \underbrace{\frac{\bar{Y}(g^d_{t-1}) + \bar{Y}(g^d_{t-1}-1)}{2}}_{\substack{\text{Average income across}\\ \text{deciles } g^d_{t-1} \text{ and } g^d_{t-1}-1}},\end{aligned} \tag{29}$$

where $\sigma^2_{i,t} = Var(\epsilon_{i,t}\,|Y_{i,t-1})$,[49] while $Var\left(\epsilon_{i,t}\,|g^d_{t-1}\right)$ and $Var\left(\epsilon_{i,t}\,|g^d_{t-1}-1\right)$ denote the time t variances, respectively, in the lagged earnings deciles g^d_{t-1} and $g^d_{t-1}-1$. The denominator $\bar{Y}(g^d_{t-1}) - \bar{Y}(g^d_{t-1}-1)$ is the difference between average earnings in deciles g^d_{t-1} and $g^d_{t-1}-1$ in year $t-1$. Therefore, the first part of expression (29) is the *difference quotient* of the conditional variance function within a given quintile g_{t-1} of the lagged earnings distribution. The second part of (29) represents a normalization to express the change in earnings across the two deciles as a share of

[47] See Table B-3.
[48] See Appendix B.2 for details.
[49] Similar to the definition in Assumption 2.

average earnings over the two deciles.

Second, to obtain the estimates of the individual risk effects (ψ_g) we average the estimated marginal changes across time:

$$\psi_g = \frac{1}{T}\sum_{t=1}^{T}\psi_{g,t}. \tag{30}$$

Figure 5 displays the value of ψ_g for each lagged earnings quintile g (red solid line, right axis). The latter is negative between the 1st and the 4th quintile and positive for the 5th quintile. In addition, ψ_g is increasing in the earnings quintiles: hence individuals who belong to higher earnings groups experience smaller (in absolute value) marginal changes in their income volatility within their earnings group.

These effects are also meaningful in terms of magnitude. Consider, for instance the first quintile of the lagged earnings distribution. The earnings volatility decreases, on average, by 0.005 units out of a 1 percent increase in lagged earnings within that quintile. In relative terms, this corresponds to an almost 1 percent decline in the variance (relative to its value in the first quintile).[50] In other words, within the first quintile, an increase in earnings of 1% (relative to average earnings in that quintile) decreases earnings volatility by almost 1% (relative to the average variance in that quintile).

4.3 Correlation between individual income cyclicalities and risk effects

Figure 5 clearly points to a negative correlation between the individual earnings sensitivity and the individual risk effect (the Pearson correlation coefficient is -0.83). Within our model, this negative sign induces an amplification of aggregate demand shocks, exacerbates the forward guidance puzzle and makes the Taylor principle insufficient for local determinacy.

In fact, the individual earnings sensitivity captures the exposure of individuals to aggregate risk (as it describes how sensitive individual income is to aggregate output). On the other hand, the individual risk effect determines the exposure of agents to their own idiosyncratic risk (as they describe how sensitive their earnings volatility is to individual earnings shocks). A negative correlation between these two effects implies that those agents who are more exposed to aggregate risk (i.e., who feature higher earnings sensitivities) are also more exposed to their individual risk (i.e., they have "more" negative individual risk effects). According to our estimates, agents in the bottom part of the earnings distribution are also more exposed to *both* aggregate and individual risk. In other words, poorer agents are those whose income decreases the most when a negative aggregate shock hits the economy and, at the same time, those whose risk (as captured by the variance in log earnings) increases the most after a decrease in earnings.

Estimation of sufficient statistics. We now focus on the estimation of the sufficient statistics in Proposition 3. To obtain average earnings sensitivites and risk effects we compute the arithmetic

[50] The conditional earnings variance in the first quintile is 0.58.

means of ψ_g and β_g across different lagged bins. The risk-cyclicality covariance effect is, instead, the sample variance of ψ_g and β_g.

$$\text{Average earnings sensitivities} = \frac{1}{5}\sum_{g=1}^{5}\psi_g,$$

$$\text{Average risk effects} = \frac{1}{5}\sum_{g=1}^{5}\beta_g,$$

$$\text{Risk-cyclicality covariance effect} = \frac{1}{5}\sum_{g=1}^{5}\psi_g\cdot\beta_g - \left(\frac{1}{5}\sum_{g=1}^{5}\psi_g\right)\cdot\left(\frac{1}{5}\sum_{g=1}^{5}\beta_g\right).$$

Finally, we estimate the parameter Γ in (18). This parameter is key as it captures the effects of heteroskedasticity. Recall that when $\Gamma = 0$ our model is isomorphic to a homoskedastic New Keynesian model. Γ is obtained as the mixed moment of ψ_g and β_g (i.e., the sum of the risk-cyclicality covariance and the product between average earnings sensitivities and risk effects):

$$\Gamma = \frac{1}{5}\sum_{g=1}^{5}\psi_g\cdot\beta_g.$$

Table 1 shows the estimates for the sufficient statistics and for the parameter Γ. We find that, on average, individual risk is decreasing over the earnings distribution (-0.13) and that agents who are more exposed to aggregate output (i.e., have higher earnings sensitivities) experience a stronger reduction in their income volatility (the covariance is negative and equal to -0.09). Overall, we find that $\Gamma = -0.27$. In our model, amplification of demand shocks is governed by compounding/discounting effects in the parameter Θ (see equation 19a), that is a decreasing function of Γ. Since $\Gamma < 1 \iff \Theta > 1$, this negative sign generates AD-compounding and, in turn, amplification of demand shocks.

Table 1: Estimates of sufficient statistics

Statistic	**Estimate**
Average individual risk effect	-.13
Average individual earnings sensitivities	1.343
Risk-cyclicality covariance effect	-.093
Parameter Γ	-.268

Notes. The table shows our estimates of the sufficient statistics in Proposition 3 and of the parameter Γ.

Hand-to-mouth agents. In an extension of the baseline model in Appendix C, we include hand-to-mouth (HtM) agents, i.e., agents who consume, in each period, their disposable income (and have hence an MPC equal to unity). In our baseline model, the average MPC in the steady state is equal to μ, which, according to empirically reasonable calibrations, is small, since it coincides with the MPC of a permanent-income consumer. When a fraction $\lambda \in (0,1)$ of HtM agents is included, we are able to recover an average MPC closer to empirical estimates (see Appendix C).

In the setting with HtM, the parameter Θ can be expressed as:

$$\Theta = \left(1 + \Xi \cdot \frac{\gamma\mu^2}{2}\Gamma\right)^{-1}$$

where:

$$\Xi := \frac{1-\lambda}{1-\lambda\cdot\chi}$$

and χ is the average elasticity of HtM income to aggregate output (following Bilbiie, 2020). In other words, the multiplier Θ is a function of three components that govern the amplification/dampening effects of fiscal and monetary policy. First, our key parameter Γ, that is a function of our sufficient statistics and captures the role of heteroskedasticity. Second, the product $\gamma\mu^2/2$, that governs the pass-through from income to consumption risk (for non-HtM agents, who are on the Euler equation). Lastly, the term Ξ, that captures the effects of fiscal and monetary policy through the presence of HtMs. In particular, notice that $\Xi > 1 \iff \chi > 1$ (assuming $\lambda < \chi^{-1}$): when HtMs have more cyclical income (relative to non-HtMs) output is more responsive to demand shocks, since HtM agents also have higher MPCs (equal to 1).

5 Conclusion

We have studied the implications of cross-sectional heteroskedasticity in individual income in a heterogeneous agent New Keynesian model. Our setup is grounded in the empirical evidence from micro data showing significant heterogeneity in the volatility of individual income over the earnings distribution.

We have assumed the existence of a structural relation linking earnings volatility to lagged income. In this setting, changes in earnings affect idiosyncratic risk and, in turn, consumption choices. Unlike a CARA-utility framework with homoskedastic income (where the MPC is uniform across agents), our model is able to generate heterogeneity in MPCs out of transitory income shocks, and to account for idiosyncratic risk and market incompleteness in an analytically tractable way. Since in our model marginal changes in individual earnings shape income volatility at the margin, saving and consumption choices are also affected at the margin, implying that individual MPCs depend on individual income.

We have shown that, through indirect general equilibrium effects, heteroskedasticity affects the response of output to demand shocks, the transmission of fiscal and monetary policy, the (in)determinacy of the equilibrium, and the FG puzzle. Those general equilibrium effects hinge on the interaction between the income distribution and changes in individual risk. Most importantly, those effects can be captured by a set of sufficient statistics which allow, in principle, to further discipline the model via a small set of moments observable in micro data.

In the data, we estimate that individual earnings sensitivities correlate negatively with marginal changes in earnings volatilities over the lagged earnings distribution. In other words, agents with

lower income, who feature more cyclical earnings, also exhibit a stronger marginal reduction in risk out of an increase in their earnings. This mechanism is at the heart of the amplification of demand shocks highlighted in the paper.

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APPENDIX

A A heteroskedastic New Keynesian model

This appendix contains the proofs of the results and the description of the supply-side of the heteroskedastic New Keynesian model of Section 3.

A.1 Proofs of Propositions 1, 2, 3

A.1.1 Proposition 1

Proof. Suppose that $c_{i,t}$ follows the process:

$$c_{i,t+1} = \phi_{i,t} c_{i,t} + \Gamma_{i,t} + v_{i,t+1}, \tag{A-1}$$

where $\phi_{i,t}$ and the drift $\Gamma_{i,t}$ are time- and individual-specific coefficients, and $v_{i,t+1}$ is the consumption process innovation in $t+1$.

- Firstly, substituting the guess (A-1) in the Euler equation (3) and taking the logs we obtain:

$$-\gamma(1-\phi_{i,t})c_{i,t} = \log[\beta(1+r_t)] - \gamma\Gamma_{i,t} + \log \mathbb{E}_t[\exp\{-\gamma v_{i,t+1}\}].$$

  Then, $\phi_{i,t} = 1$. Otherwise consumption would be entirely determined from the Euler equation, without any reference to the intertemporal budget constraint (see Caballero, 1990). Imposing $\phi_{i,t} = 1$, we can write the drift as a function of the real interest rate and the moments of the consumption process innovation:

$$\Gamma_{i,t} = \frac{1}{\gamma}\log[\beta(1+r_t)] + \frac{1}{\gamma}\log \mathbb{E}_t[\exp\{-\gamma v_{i,t+1}\}]. \tag{A-2}$$

- Secondly, let us consider the intertemporal budget constraint (iBC):

$$\sum_{k\geq 0} Q_{t,t+k} c_{i,t+k} = \sum_{k\geq 0} Q_{t,t+k} y_{i,t+k} + a_{i,t}, \tag{A-3}$$

  where $Q_{t,t+k} := [(1+r_t)\cdot \ldots \cdot (1+r_{t+k-1})]^{-1}$ for $k \geq 1$ and $Q_{t,t} = 1$. Additionally, let us integrate backward the consumption process in (A-1) (with $\phi_{i,t} = 1$):

$$c_{i,t+k} = c_{i,t} + \sum_{j=1}^{k}(\Gamma_{i,t+j-1} + v_{i,t+j}),$$

Substituting this expression in the iBC, after some manipulations, we obtain:

$$\begin{aligned}&\sum_{k\geq 0} Q_{t,t+k} c_{i,t} + \sum_{k\geq 1} Q_{t,t+k} \sum_{j=1}^{k} (\Gamma_{i,t+j-1} + v_{i,t+j}) = \\ &= a_{i,t} + \sum_{k\geq 0} Q_{t,t+k} \mathbb{E}_t(y_{i,t+k}) + \sum_{k\geq 1} Q_{t,t+k} \underbrace{[y_{i,t+k} - \mathbb{E}_t(y_{i,t+k})]}_{\epsilon_{i,t+k}}.\end{aligned} \tag{A-4}$$

- Taking the expectations of (A-4) conditional on time t information, we obtain the solution for consumption:

$$c_{i,t} = \mu_t a_{i,t} + \mu_t \left(y_{i,t} + \sum_{k\geq 1} Q_{t,t+k} y^d_{i,t+k} \right) - \tilde{\Gamma}_{i,t}, \tag{A-5}$$

  where μ_t is the MPC out of wealth, such that: $\mu_t := \left(\sum_{k\geq 0} Q_{t,t+k} \right)^{-1}$ and the constant $\tilde{\Gamma}_{i,t}$ is defined as:

$$\tilde{\Gamma}_{i,t} := \mu_t \sum_{k\geq 1} Q_{t,t+k} \sum_{j=1}^{k} \Gamma_{i,t+j-1}.$$

  Note that we can establish (as in Acharya and Dogra, 2020) the following recursion for the wealth MPC:

$$\mu_t^{-1} = 1 + Q_{t,t+1} \mu_{t+1}^{-1} \iff \mu_t = \frac{\mu_{t+1}(1+r_t)}{1+\mu_{t+1}(1+r_t)}.$$

- To solve for the consumption process innovation as a function of the structural income innovation, we can combine equations (A-4) and (A-5). Then:[51]

$$\begin{aligned}&\sum_{k\geq 1} Q_{t,t+k} \left(\sum_{j=1}^{k} v_{i,t+j} - \epsilon_{i,t+k} \right) = 0 \iff {}^{52} \\ &\sum_{j\geq k} Q_{t,t+j} v_{i,t+k} - Q_{t,t+k} \epsilon_{i,t+k} = 0 \quad \forall k \geq 1 \implies\end{aligned}$$

$$v_{i,t+k} = \mu_{t+k} \epsilon_{i,t+k} \quad \forall k \geq 1.^{53} \tag{A-6}$$

[51] Note that the following result holds since we assumed that the entire path of variances is known (given the dependence on the deterministic components of lagged income). Otherwise, if the path of variances were stochastic, equation (A-5) would have to be modified to:

$$c_{i,t} = \mu_t a_{i,t} + \mu_t \left(y_{i,t} + \sum_{k\geq 1} Q_{t,t+k} y^d_{i,t+k} \right) - \mathbb{E}_t(\tilde{\Gamma}_{i,t}).$$

Then, combining the previous equation with equation (A-4), we would get, almost surely:

$$\sum_{k\geq 1} Q_{t,t+k} \left(\sum_{j=1}^{k} v_{i,t+j} - \epsilon_{i,t+k} \right) + \sum_{k\geq 1} Q_{t,t+k} \sum_{j=1}^{k} \left[\Gamma_{i,t+j-1} - \mathbb{E}_t(\Gamma_{i,t+j-1}) \right] = 0,$$

where $\Gamma_{i,t+j-1}$ depends on the moments of the income innovation process whose variance is stochastic.

[52] While the *if* part of this statement is trivial, the *only if* can be proved by contraposition, exploiting the time-independence of consumption and income innovations.

Substituting equation (A-6) back in equation (A-2) and using the moments of the log-Normal distribution,[54] we can finally obtain the formula for the drift of the consumption process.

$$\Gamma_{i,t} = \frac{1}{\gamma}\log[\beta(1+r_t)] + \frac{\gamma\mu_{t+1}^2}{2}\sigma_{i,t+1}^2. \tag{A-7}$$

To conclude, substituting the drift (A-7) and the consumption innovation (A-6) back in the initial guess, we obtain the stochastic difference equation that characterize the solution for the consumption process.

$$c_{i,t} = c_{i,t+1} \underbrace{-\frac{1}{\gamma}\log[\beta(1+r_t)] - \frac{\gamma\mu_{t+1}^2}{2}\sigma_{i,t+1}^2}_{=-\Gamma_{i,t}} \underbrace{-\mu_{t+1}\epsilon_{i,t+1}}_{=-v_{i,t+1}}.$$

■

A.1.2 Proposition 2

Proof. Let us define the MPC out of income as the partial derivative of the optimal consumption function to current income: $MPC_{i,t} := \frac{\partial c_{i,t}}{\partial y_{i,t}}$. To compute equation (6), we can differentiate equation (A-5):

$$\begin{aligned} MPC_{i,t} := \frac{\partial c_{i,t}}{\partial y_{i,t}} =& \mu_t - \frac{\partial \tilde{\Gamma}_{i,t}}{\partial y_{i,t}} = \\ =& \mu_t - \mu_t \sum_{k\geq 1} Q_{t,t+k}\frac{\partial \Gamma_{i,t}}{\partial y_{i,t}} = \\ =& \mu_t - \frac{\mu_t}{(1+r_t)\mu_{t+1}}\frac{\gamma\mu_{t+1}^2}{2}\frac{\partial \sigma_{i,t+1}^2}{\partial y_{i,t}} = \\ =& \mu_t - \frac{1}{1+\mu_{t+1}(1+r_t)}\frac{\gamma\mu_{t+1}^2}{2}\frac{\partial \sigma_{i,t+1}^2}{\partial y_{i,t}}. \end{aligned}$$

It is clear that if $\partial\sigma_{i,t+1}^2/\partial y_{i,t} \neq 0$, then the MPC out of wealth differs from the one out of transitory income: $\mu_t \neq MPC_{i,t}$. Moreover, if $\partial\sigma_{i,t+1}^2/\partial y_{i,t}$ is non-constant with respect to $y_{i,t}$, then the distribution of MPCs is non-degenerate.

■

A.1.3 Proposition 3

Proof. Consider a fully transitory government spending shock g_t (i.e., $g_t \neq 0$, $g_\tau = 0$ for $\tau > t$) and let the monetary authority follow the interest rate peg rule (i.e., $r_\tau = r$, with $v_\tau = 0$ for every $\tau \geq t$). Let

[53] Since $Q_{t,t+k}(\sum_{j\geq k} Q_{t,t+j})^{-1} = \mu_{t+k}$.

[54] Since $-\gamma\mu_{t+1}\epsilon_{i,t+1}|\mathcal{I}_t \sim \mathcal{N}(0, \gamma^2\mu_{t+1}^2\sigma_{i,t+1}^2)$, then $\mathbb{E}_t\big[\exp\{-\gamma\mu_{t+1}\epsilon_{i,t+1}\}\big] = \exp\left\{\frac{\gamma^2\mu_{t+1}^2}{2}\sigma_{i,t+1}^2\right\}$.

$\mathcal{H} : \mathbb{R}^2 \to \mathbb{R}$ be the function linking aggregate output and spending (for a given $y_{t+1}, g_{t+1}, r_t, \mu_{t+1}$), defined by:

$$\mathcal{H}(y_t, g_t) = y_t - y_{t+1} + \frac{1}{\gamma}\log[\beta(1+r_t)] + \frac{\gamma\mu_{t+1}^2}{2}\int_{[0,1]}\sigma_{i,t+1}^2 di - g_t + g_{t+1}.$$

Notice that at the solution (y_t, g_t) to the equation $\mathcal{H}(y_t, g_t) = 0$, the dynamic AD equation (9) holds. By the Implicit Function Theorem, if $\mathcal{H} \in \mathcal{C}^1$ (space of continuously differentiable functions) and $\mathcal{H}_y(y_t, g_t) \neq 0$,[55] then there exist two neighborhoods $\mathcal{U}(y_t)$ and $\mathcal{V}(g_t)$ and a function $\tilde{F} : \mathcal{V}(g_t) \to \mathcal{U}(y_t)$ such that:

$$y_t = \tilde{F}(g_t) \iff \mathcal{H}(y_t, g_t) = 0 \quad \forall g_t \in \mathcal{V}(g_t),$$

and:

$$\frac{\partial y_t}{\partial g_t} = \tilde{F}'(g_t) = -\frac{\mathcal{H}_g(y_t, g_t)}{\mathcal{H}_y(y_t, g_t)}.$$

Since:

$$\begin{aligned}
\mathcal{H}_g(y_t, g_t) &= -1, \\
\mathcal{H}_y(y_t, g_t) &= 1 + \frac{\gamma\mu_{t+1}^2}{2}\frac{d}{dy_t}\int_{[0,1]}\sigma_{i,t+1}^2\ di,
\end{aligned}$$

and assuming that we can exchange the integral and the derivative:[56]

$$\begin{aligned}
\frac{d}{dy_t}\int_{[0,1]}\sigma_{i,t+1}^2\ di &= \int_{[0,1]}\left(\frac{\partial\sigma_{i,t+1}^2}{\partial y_{i,t}}\frac{\partial y_{i,t}}{\partial y_t}\right)di = \\
&= \int_{[0,1]}\frac{\partial\sigma_{i,t+1}^2}{\partial y_{i,t}}di\int_{[0,1]}\frac{\partial y_{i,t}}{\partial y_t}di + Cov\left(\frac{\partial\sigma_{i,t+1}^2}{\partial y_{i,t}}, \frac{\partial y_{i,t}}{\partial y_t}\right),
\end{aligned}$$

we obtain the output multiplier of government spending:

$$\Theta_t := \frac{\partial y_t}{\partial g_t} = \left\{1 + \frac{\gamma\mu_{t+1}^2}{2}\left[\mathbb{E}\left(\frac{\partial\sigma_{i,t+1}^2}{\partial y_{i,t}}\right)\cdot\mathbb{E}\left(\frac{\partial y_{i,t}}{\partial y_t}\right) + Cov\left(\frac{\partial\sigma_{i,t+1}^2}{\partial y_{i,t}}, \frac{\partial y_{i,t}}{\partial y_t}\right)\right]\right\}^{-1}.$$

Following the same steps, we can derive the output multiplier out of a fully transitory interest rate shock:[57]

$$\frac{\partial y_t}{\partial r_t} = -\Theta_t \cdot \frac{1}{\gamma(1+r_t)}. \tag{A-8}$$

To conclude, since Θ_t depends on the three cross-sectional statistics described in Proposition 3, we can state that these are *sufficient* to characterize the IRFs of aggregate output to demand shocks.

■

[55] With the notation H_y we mean the partial derivative of H to y.

[56] It is sufficient to assume that the function $\sigma_{i,t+1}^2 = \sigma^2(y_{i,t}) = \sigma^2(i, y_t)$ (the last equality follows from $y_{i,t} = y(i, y_t)$) is continuously differentiable.

[57] Equivalently, a shock to v_t.

A.2 Income process, supply-side, and government

We model the supply-side and the government-side of our heteroskedastic economy following Acharya and Dogra (2020).

A.2.1 Income process

We assume that labor productivity is the underlying source of households' idiosyncratic risk. In particular, this structure allows to model the income process to incorporate: (1) the presence of a permanent component of earnings; (2) the empirical fact that the elasticity of individual to aggregate earnings is heterogeneous over the earnings distribution (Guvenen et al., 2017).

We assume that income $y_{i,t}$ is given by the sum of labor earnings, dividends ($d_{i,t}$), and government transfers ($T_{i,t}$). Labor earnings are, in turn, equal to the real wage (w_t) net of taxes (where τ_t is the tax rate) times the total number of hours worked. In particular, total hours are represented by the sum of an agent-specific productivity factor $l^d_{i,t}$ (that evolves deterministically over time) and of a stochastic component $l_{i,t}$.

We assume that the stochastic component of labor productivity $l_{i,t}$ follows a zero-mean Gaussian distribution, conditional on the previous period information set: $l_{i,t} \mid \mathcal{I}_{t-1} \sim \mathcal{N}(0, \sigma^2_{l,i,t})$. The volatility of this process is heterogeneous in the cross-section and depends on lagged deterministic earnings $\sigma^2_{l,i,t} := \sigma^2(y^d_{i,t-1})$.[58] Under these assumptions, the part of labor earnings that is determined by the agent-specific productivity factor corresponds to what in Section 3 we call *deterministic income*: $y^d_{i,t} := (1-\tau_t) \cdot w_t \cdot l^d_{i,t}$. Meanwhile, the *stochastic component* is: $\epsilon_{i,t} = (1-\tau) \cdot w_t \cdot l_{i,t}$. Therefore:

$$\begin{aligned} y_{i,t} &= \underbrace{(1-\tau_t) \cdot w_t \cdot (l^d_{i,t} + l_{i,t})}_{\text{labor earnings}} + d_{i,t} + T_{i,t} = \\ &= \underbrace{(1-\tau_t) \cdot w_t \cdot l^d_{i,t}}_{y^d_{i,t}} + \underbrace{(1-\tau) \cdot w_t \cdot l_{i,t}}_{\epsilon_{i,t}} + d_{i,t} + T_{i,t}. \end{aligned} \tag{A-9}$$

Finally, we assume that, in every period, average hours are equal to 1: $\int_{[0,1]} l^d_{i,t}\, di = 1$. Under this specification for the income process we can make some considerations.

- The presence of the factor $l^d_{i,t}$ captures the permanent component of earnings. Suppose, henceforth, that $l^d_{i,t} = H(y_t, l^d_{i,-1}) = l^d_{i,-1} + h_i(y_t)$ where h_i is an individual-specific function of aggregate output. Under this formulation, $l^d_{i,t}$ is decomposed into an individual productivity *fixed effect* $l^d_{i,-1}$ and a time effect $h_i(y_t)$ (through the function h_i).

- Since that $l^d_{i,t}$ changes over the business cycle, parametrizing $l^d_{i,t}$ as an appropriate function of y_t (i.e., parametrizing the function h_i), we are able to reproduce the *worker betas* (Guvenen

[58] This assumption captures the idea that, for example, richer agents might be exposed to a full set of experiences and opportunities that make them more certain about their future labor productivity.

et al., 2017), i.e., the elasticities of individual to aggregate earnings. Guvenen et al. (2017) show that *worker betas* are heterogeneous and U-shaped over the earnings distribution. In our setting, we can compute *worker betas* as:[59]

$$\beta_{i,w} = \frac{\partial y_{i,t}^{labor}}{\partial y_t} \cdot \frac{y_t}{y_{i,t}^{labor}} = \frac{\partial w_t}{\partial y_t} \cdot \frac{y_t}{w_t} + \frac{\partial l_{i,t}^d}{\partial y_t} \cdot \frac{y_t}{l_{i,t}^d + l_{i,t}}.$$

- Our income process is akin to a traditional *permanent/transitory* structural income process. The transitory component coincides with our stochastic component $\epsilon_{i,t}$ and the permanent one coincides with our deterministic component $y_{i,t}^d$, that is characterized by the presence of the fixed effect $l_{i,-1}^d$.

To simplify the analysis, assume henceforth uniform taxes and dividends: $\tau_t = \tau$, $d_{i,t} = d_t$, and $T_{i,t} = T_t$.

A.2.2 Technology

There is a continuum of intermediate firms, of measure 1, that produce a variety j of the intermediate good $x_t(j)$, according to the constant return to scale technology:

$$x_t(j) = z_t m_t(j)^{\alpha} n_t(j)^{1-\alpha}, \tag{A-10}$$

with $\alpha \in (0,1)$. $m_t(j) = (\int_{[0,1]} m_t(j,k)^{\frac{\epsilon-1}{\epsilon}} dk)^{\frac{\epsilon}{\epsilon-1}}$ is the Dixit and Stiglitz (1977) aggregator of the varieties of the intermediate goods that are re-used in the production process ($m_t(j,k)$ represents the variety k purchased by the firm j). Labor employed by firm j is denoted by $n_t(j)$ and z_t is the common total factor productivity (TFP).

Given the constant elasticity of substitution (CES) aggregator for the intermediate good m_t, the demand function for variety k by firm j is:

$$m_t(j,k) = \left(\frac{p_t(k)}{p_t}\right)^{-\epsilon} m_t(j),$$

with $p_t := (\int_{[0,1]} p_t(k)^{1-\epsilon} dk)^{\frac{1}{1-\epsilon}}$ being the aggregate price index. Then, total demand for variety k of the intermediate good is given by:

$$m_t(k) := \int_{[0,1]} m_t(j,k) dj = \left(\frac{p_t(k)}{p_t}\right)^{-\epsilon} \underbrace{\int_{[0,1]} m_t(j) dj}_{m_t :=} = \left(\frac{p_t(k)}{p_t}\right)^{-\epsilon} m_t.$$

Finally, there is a perfectly competitive firm producing a final good y_t according to the CES technology $y_t = (\int_{[0,1]} y_t(k)^{\frac{\epsilon-1}{\epsilon}} dk)^{\frac{\epsilon}{\epsilon-1}}$. Then, the demand function from the final good producer for

[59] Let us ignore dividends and transfers and consider only labor earnings $y_{i,t}^{labor} = (1-\tau_t) \cdot w_t \cdot (l_{i,t}^d + l_{i,t})$.

each variety k is:

$$y_t(k) = \left(\frac{p_t(k)}{p_t}\right)^{-\epsilon} y_t.$$

Summarizing, good $x_t(k)$ (produced by the intermediate good firm k) is purchased in part by other intermediate good firms ($m_t(k)$), and in part by the final good firm ($y_t(k)$):

$$x_t(k) = m_t(k) + y_t(k) = \left(\frac{p_t(k)}{p_t}\right)^{-\epsilon} \underbrace{(m_t + y_t)}_{x_t :=} = \left(\frac{p_t(k)}{p_t}\right)^{-\epsilon} x_t. \tag{A-11}$$

Therefore, y_t can be also interpreted as net output: $y_t = x_t - m_t$.

A.2.3 Cost minimization

Each intermediate producer purchases the aggregated intermediate goods and hires workers, solving the cost minimization problem (expressed in real terms):

$$\begin{aligned} \mathcal{C}_t(x_t(j)) := \min_{(m_t(j), n_t(j))} & \left[\frac{p_t(j)}{p_t} m_t(j) + w_t n_t(j)\right] \\ s.t. \qquad & x_t(j) \leq z_t m_t(j)^{\alpha} n_t(j)^{1-\alpha}. \end{aligned} \tag{A-12}$$

We get the following optimality conditions, considering a symmetric equilibrium[60] and imposing the labor market clearing, i.e., $\int_{[0,1]} n_t(j)dj = \int_{[0,1]} (l^d_{i,t} + l_{i,t})di = 1$:

$$x_t = z_t m_t^{\alpha}, \tag{A-13a}$$

$$w_t = \frac{1-\alpha}{\alpha}\left(\frac{x_t}{z_t}\right)^{\frac{1}{\alpha}}, \tag{A-13b}$$

$$\mathcal{C}_t(x_t) = \frac{1}{\alpha}\left(\frac{x_t}{z_t}\right)^{\frac{1}{\alpha}}, \tag{A-13c}$$

where (A-13a) is the production function, (A-13b) is the optimality condition (MRTS = real wage) and (A-13c) is the (real) cost function.

Finally, we can determine an equation linking aggregate output and the real wage, by combining equations (A-13a), (A-13b) and the definition of net output.

$$y_t = z_t \cdot \left(\frac{\alpha}{1-\alpha}\right)^{\alpha} \cdot w_t^{\alpha} - \frac{\alpha}{1-\alpha} \cdot w_t. \tag{A-14}$$

We can, then, determine aggregate real dividends:

$$d_t = x_t - \mathcal{C}_t(x_t) = x_t - m_t - w_t = y_t - w_t. \tag{A-15}$$

[60] The symmetry is implied by the price stickiness à la Rotemberg (1982), that we assume here. See Section A.2.4.

A.2.4 Nominal frictions

We model nominal rigidities following Rotemberg (1982). Each intermediate good firm sets prices in every period paying a quadratic convex adjustment cost (rebated lump-sum and uniformly to households)[61] and solves the following recursive maximization problem:

$$V_t^f(p_{t-1}(j)) = \max_{p_t(j)} \left\{ \left[\frac{p_t(j)}{p_t} x_t(j) - \mathcal{C}_t(x_t(j)) - \frac{\psi}{2}\left(\frac{p_t(j)}{p_{t-1}(j)} - 1\right)^2 x_t \right] + \frac{1}{1+r_t} V_{t+1}^f(p_t(j)) \right\}$$
$$s.t \quad x_t(j) = \left(\frac{p_t(j)}{p_t}\right)^{-\epsilon} x_t,$$

where ψ regulates the size of the adjustment cost (hence, the degree of price rigidity in the economy) and $V_t^f(\cdot)$ is the value function of the firm optimization problem.

From the optimality conditions, exploiting the symmetry of the equilibrium, we obtain the non-linear PC:

$$\psi(1+\pi_t)\pi_t = 1 - \epsilon(1 - \mathcal{MC}_t(x_t)) + \psi \cdot \frac{1}{1+r_t}(1+\pi_{t+1})\pi_{t+1}\frac{x_{t+1}}{x_t}, \tag{A-16}$$

where $\pi_t := \frac{p_t}{p_{t-1}} - 1$ and $\mathcal{MC}_t(x_t) := \frac{\partial \mathcal{C}_t}{\partial x_t} = \alpha^{-2} z_t^{-1/\alpha} x_t^{\frac{1-\alpha}{\alpha}}$.

Linearizing equation (A-16) around the zero-inflation steady state (setting $z_t = z$ and normalizing $y = 1$) described in Section 3.5.1, we obtain the linearized PC of equation (19c):

$$\hat{\pi}_t = \tilde{\beta} \cdot \hat{\pi}_{t+1} + k \cdot \hat{y}_t,$$

where the *hat* notation denotes variables in deviation from their steady-state value, $\tilde{\beta} := \frac{1}{1+r}$, and the slope of the PC is:[62]

$$k := \frac{\epsilon}{\psi}\frac{1-\alpha}{\alpha^2}\frac{1}{x}\left(\alpha z^{\frac{1}{\alpha}} x^{1-\frac{1}{\alpha}} - 1\right)^{-1}.$$

[61] Following Ascari and Rossi (2012), footnote 17, this means that the true households' budget constraint should be:

$$c_{i,t} + \frac{a_{i,t+1}}{1+r_t} = y_{i,t} + a_{i,t} + \frac{\psi}{2}\left(\frac{p_t(j)}{p_{t-1}(j)} - 1\right)^2 x_t.$$

This assumption implies that in equilibrium there is no waste from price adjustment costs and $y_t = c_t + g_t$.

[62] This expression can be obtained by combining the linearized versions of equations (A-13a) and $y_t = x_t - m_t$ (since $\hat{y}_t = x \cdot \hat{x}_t - m \cdot \hat{m}_t$ and $\hat{x}_t = \alpha \cdot \hat{m}_t$, with the steady-state value of x solving $x - (x/z)^{1/\alpha} = 1$):

$$\hat{x}_t = \left[x - \frac{1}{\alpha}\left(\frac{x}{z}\right)^{\frac{1}{\alpha}}\right]^{-1} \hat{y}_t.$$

The expression above can then be used to substitute $\hat{x}_t$ with $\hat{y}_t$ in:

$$\frac{\epsilon}{\psi}\mathcal{MC}_t'(x) \cdot x \cdot \hat{x}_t = \frac{\epsilon}{\psi}\frac{1-\alpha}{\alpha^3}\frac{1}{z^{\frac{1}{\alpha}}} x^{\frac{1-\alpha}{\alpha}} \hat{x}_t.$$

A.2.5 Government

The government purchases the final good as g_t and pays lump-sum transfers T_t to households. In addition, it can raise proportional labor income taxes $\tau_t w_t$ and finances its deficits with government debt (in real terms) b_t. The intra-temporal government budget constraint is:

$$\frac{b_{t+1}}{1+r_t} = b_t - s_t, \tag{A-17}$$

where s_t is primary surplus:

$$s_t = \tau_t w_t - g_t - T_t. \tag{A-18}$$

A.2.6 Equilibrium

Markets clear in general equilibrium.

- Financial market clearing: $a_t = \int_{[0,1]} a_{i,t}\ di = b_t$.
- Labor market clearing: $n_t = \int_{[0,1]} n_t(j)\ dj = \int_{[0,1]} (l^d_{i,t} + l_{i,t})\ di = 1$.
- Goods market clearing for each intermediate good: $x_t(k) = y_t(k) + m_t(k) \implies x_t = y_t + m_t$.
- Final good market clearing: $y_t = c_t + g_t$ where $c_t = \int_{[0,1]} c_{i,t}\ di$.

Notice that we are integrating with respect to the Lebesgue measure. In particular, let $\bar{y}_t$ be average household earnings, defined as $\bar{y}_t := \int_{[0,1]} y_{i,t}\ di = (1-\tau_t)w_t + d_t + T_t$[63], then (exploiting the definition of $\bar{y}_t$, d_t and the financial market clearing) we can prove that the following relations hold in general equilibrium:

$$\bar{y}_t + s_t = w_t + d_t - g_t,$$
$$y_t = s_t + \bar{y}_t + g_t.$$

A.3 Steady state

In this section we prove the statements of Section 3.5.1, where we describe the steady state of the model. Recall that we set $g = 0$ and normalize aggregate consumption and output to 1 in the steady state. The real wage is determined from equation (A-14),[64] given the parameters $\alpha \in (0,1), z > 0$. In addition, the deterministic component of labor productivity follows: $l^d_{i,t} = l^d_{i,-1} + h_i(y_t)$ with $h_i(y) = 0$ (where y is steady-state output).

Lemma A.1. *Let us assume that the initial fixed effects $l^d_{i,-1}$ are i.i.d. draws from the cumulative distribution function (CDF) $\bar{F} : \mathbb{R} \to [0,1]$. Then, $F : \mathbb{R} \to [0,1]$, the CDF describing the stationary distribution of (deterministic) income, exists in the steady state.*

[63] Since, by a continuous version of the LLN, $\int_{[0,1]} l_{i,t}\ di = 0$. See Uhlig (1988), Proposition 1.

[64] Notice that this equation admits more than one solution. We consider the lowest possible solution for the real wage.

Proof. Let $y = 1$ and the real wage w be determined from (A-14). The expression for deterministic income is:

$$y_i^d = (1-\tau)w \cdot l_i^d = (1-\tau)w \cdot l_{i,-1}^d,$$

since in the steady state we take the permanent component of labor productivity to be equal to the initial extraction of the fixed effect: $l_i^d = l_{i,-1}^d$. Since y_i^d is a scalar transformation of $l_{i,-1}^d$, for every $l_{i,-1}^d$, let $F : \mathbb{R} \to [0,1]$ be defined as:

$$F(y_i^d) = \bar{F}(l_{i,-1}^d) \quad \text{with} \quad y_i^d = (1-\tau)w \cdot l_{i,-1}^d.$$

Then, since $\bar{F}$ is right-continuous and non-decreasing (as it is a CDF) and y_i^d is a linear transformation of $l_{i,-1}^d$, then F is also right-continuous and non-decreasing. We can then argue (by Carathéodory extension theorem) that F is the CDF of a unique probability measure on $\mathbb{R}$, describing the distribution of deterministic income.

■

Lemma A.2. *Consider the measure space* $([0,1], \mathcal{B}([0,1]), \nu)$*, where* ν *is the Lebesgue measure, and the measurable space* $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ *where* $\mathcal{B}$ *denotes the Borel sigma-algebra. Let us assume that the function* $y^d : [0,1] \to \mathbb{R}$ $(i \mapsto y_i^d)$ *is measurable. Then, the CDF of deterministic income,* $F : \mathbb{R} \to [0,1]$*, represents the Lebesgue-Stieltjes measure of the push-forward of* ν *through* y^d*.*

Therefore, we can equivalently compute the average volatility weighting either by the CDF of deterministic income F *or by the Lebesgue measure* ν*:*

$$\bar{\sigma}^2 := \int_{\mathbb{R}} \sigma^2(y)\ dF(y) = \int_{[0,1]} \sigma^2(y_i^d)\ \nu(di).$$

Proof. By definition[65] of *push-forward* measure,[66] that we denote by $y^d \# \nu$, we have that:

$$(y^d \# \nu)(B) = \nu\Big(\underbrace{(y^d)^{-1}(B)}_{\substack{\text{counter-image} \\ \text{of } B \text{ through } y^d}} \Big) \quad \text{for every } B \in \mathcal{B}(\mathbb{R}).$$

Notice that $y^d \# \nu$ is a measure on the target space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We want to show that this measure admits a unique CDF representation, coinciding with F. Consider the *semi-algebra* defined by the semi-intervals of the kind $(-\infty, y]$. Then, we have:

[65] Since the function y^d is measurable, the definition below is meaningful.

[66] The push-forward measure allows to define a measure on a target measurable space – here: $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ – starting from a measure on an original space – here: ν on $([0,1], \mathcal{B}([0,1]))$ – through a function that maps from the original to the target space – here: y^d.

$$
\begin{aligned}
(y^d \# \nu)\left((-\infty, y]\right) =& \nu\Big((y^d)^{-1}((-\infty, y])\Big) = \\
=& \nu\Big(\{i \in [0,1] : y_i^d \leq y\}\Big) = \\
=& \int_{[0,1]} \mathbb{I}\left(y_i^d \in (-\infty, y]\right) \nu(di) = \text{[67]} \\
=& F(y).
\end{aligned}
$$

The first equality is by definition of push-forward measure and the second one is by definition of counter-image. The third one follows, instead, from the definition of the Lebesgue integral and from the fact that $\mathbb{I}\left(y_i^d \in (-\infty, y]\right) = 1 \iff i \in \{i : y_i^d \leq y\}$. Finally, the last equality comes from a continuous version of the Law of Large Number.[68] In fact, let $Z_i := \mathbb{I}\left(y_i^d \in (-\infty, y]\right)$ and notice that the variables of the collection $(Z_i)_{[0,1]}$ are identically distributed and pairwise independent, with mean $\mathbb{E}(Z_i) = \mathbb{P}\left(y_i^d \in (-\infty, y]\right) = F(y)$.

From the previous equalities, it follows that $y^d \# \nu$ is a probability measure (since $(y^d \# \nu)(\mathbb{R}) = 1$) and, on the intervals of the kind $(-\infty, y]$, it coincides with F. Then, by Carathéodory extension theorem, F must be the unique representation of $y^d \# \nu$.[69]

Finally, it can be shown that the variance function $\sigma^2 : \mathbb{R} \to \mathbb{R}_+$ is integrable with respect to F *if and only if* the composition $\sigma^2 \circ y^d : [0,1] \to R_+$ is integrable with respect to ν. Moreover:

$$
\int_{\mathbb{R}} \sigma^2(y)\ dF(y) = \int_{[0,1]} \sigma^2(y_i^d)\ \nu(di).\text{[70]}
$$

■

Lemma A.3. *Existence and uniqueness of the real interest rate. If $\bar{\sigma}^2 > 0$, there exists a unique solution $r > 0$ to equation (16).*

Proof. We rewrite here the equation determining the real interest rate in the steady state:

$$
\left(\frac{1+r}{r}\right)^2 \log[\beta \cdot (1+r)] = -\frac{\gamma^2}{2}\bar{\sigma}^2.
$$

Let $r > 0$. Notice that in the previous equation since $\bar{\sigma}^2 > 0$ then $\beta(1+r) < 1$ (a standard result in the incomplete markets literature to bound the solution to the income fluctuation problem). Then, note that the LHS of equation (16) is continuous and strictly increasing in r for $r \in (0, 1/\beta - 1]$. Moreover, since the LHS $\to -\infty$ when $r \to 0$ and LHS $\to 0$ when $r \to 1/\beta - 1$, then there exists a unique solution r to equation (16).

[67] $\mathbb{I}$ is an indicator function.

[68] See Uhlig (1988), Proposition 1.

[69] More precisely: the probability measure represented by F (that is unique) must coincide with $y^d \# \nu$. In other words, $y^d \# \nu$ is the probability measure describing the stationary distribution of deterministic income.

[70] In the paper we have denoted the integration with respect to the Lebesgue measure with di instead of $\nu(di)$.

■

A.4 Equilibrium determinacy

Assume that monetary policy follows the Taylor rule, equation (12). Rewriting the linearized model - equations (19) - in matrix form,

$$\begin{pmatrix}\hat{y}_{t+1}\\ \hat{\mu}_{t+1}\\ \hat{\pi}_{t+1}\end{pmatrix} = \underbrace{\begin{bmatrix}\Theta^{-1}+\frac{k}{\tilde{\beta}}(\gamma^{-1}-\Lambda) & \frac{\Lambda}{\tilde{\beta}} & (\gamma^{-1}-\Lambda)\left(\phi_\pi-\frac{1}{\tilde{\beta}}\right)\\ -\frac{k}{\tilde{\beta}} & \frac{1}{\tilde{\beta}} & \frac{1}{\tilde{\beta}}-\phi_\pi\\ -\frac{k}{\tilde{\beta}} & 0 & \frac{1}{\tilde{\beta}}\end{bmatrix}}_{\Omega:=}\begin{pmatrix}\hat{y}_t\\ \hat{\mu}_t\\ \hat{\pi}_t\end{pmatrix} + \begin{bmatrix}1\\0\\0\end{bmatrix}\Delta\hat{g}_{t+1} + \begin{bmatrix}\gamma^{-1}-\Lambda\\ -1\\ 0\end{bmatrix}v_t,$$

determinacy requires that the matrix Ω has all its three eigenvalues outside the unit circle since the three variables $\hat{y}_t, \hat{\mu}_t, \hat{\pi}_t$ are forward-looking. We can obtain the characteristic polynomial of the matrix Ω as $\mathcal{P}(z) = \det(\Omega - I_3 \cdot z)$.[71]

$$\mathcal{P}(z) = -z^3 + \frac{2\tilde{\beta}+\tilde{\beta}^2\Theta^{-1}+k\tilde{\beta}\cdot(\gamma^{-1}-\Lambda)}{\tilde{\beta}^2}\cdot z^2 - \\ -\frac{1+2\tilde{\beta}\Theta^{-1}+k\gamma^{-1}+\tilde{\beta}k\phi_\pi\cdot(\gamma^{-1}-\Lambda)}{\tilde{\beta}^2}\cdot z + \frac{\Theta^{-1}+k\gamma^{-1}\phi_\pi}{\tilde{\beta}^2}.$$

The eigenvalues of the matrix solve the equation $\mathcal{P}(z^*) = 0$. Necessary and sufficient conditions for equilibrium determinacy are that $|z_i^*| > 1$ for $i = 1, 2, 3$.

We can find a necessary condition for the eigenvalues to be outside the unit circle under the following parametric assumption.

Assumption 4. *Assume that $\gamma^{-1} \geq \Lambda$.*

Proposition 4. *Under assumption 4, the following condition:*

$$\phi_\pi > 1 + \frac{1}{k}\left[\frac{\gamma\cdot(1-\tilde{\beta})^2}{1-\tilde{\beta}+\tilde{\beta}\cdot\gamma\Lambda}\right]\left(1-\frac{1}{\Theta}\right) =: \bar{\phi}_\pi, \tag{A-19}$$

is necessary for equilibrium determinacy since it is necessary[72] *for the three eigenvalues to be outside the unit circle.*

Proof. The polynomial $\mathcal{P}(z)$ is continuous over the real line. Moreover, $\mathcal{P}(z) \to +\infty$ as $z \to -\infty$ and $\mathcal{P}(z) \to -\infty$ as $z \to +\infty$. Under assumption 4, for every $z \leq 0$, $\mathcal{P}(z) > 0$, therefore the root of

[71] I_3 denotes the 3×3 identity matrix.

[72] Acharya and Dogra (2020) show that under additional assumptions on the discount factor being sufficiently large and Θ not being too large, condition (A-19) is also sufficient for all the eigenvalues to be outside the unit circle. See Acharya and Dogra (2020), Appendix B.3.

the polynomial must be positive.
$\mathcal{P}(1) > 0$ is a necessary condition for the roots to be outside the unit circle. By contraposition, if $\mathcal{P}(1) \leq 0$, then either $\mathcal{P}(1) = 0$ or $\mathcal{P}(1) < 0$. In latter case, the Intermediate Value Theorem[73] implies that at least one root z_i^* must be between $(0, 1)$.
Therefore, imposing $\mathcal{P}(1) > 0$ and rearranging, we get condition (A-19).

■

Notice that when $\Theta = 1$, as in the RA and homoskedastic models, this necessary condition collapses to the well-known *Taylor principle*: $\phi_\pi > 1$. In addition, since the RHS of condition (A-19) – $\bar{\phi}_\pi$ – is increasing in Θ, the more demand shocks are amplified, the stronger the reaction of monetary policy must be to stabilize the economy.

A.5 Multipliers with persistent shocks

Applying the method of undetermined coefficient, as explained in Section 3.5.4, we can obtain the multipliers of fiscal and monetary policy when shocks are persistent. For the *fiscal shock*, we have:

$$\begin{aligned}
\psi_y^g &= \frac{\gamma\Theta \cdot (1-\rho_g) \cdot (1-\tilde{\beta}\rho_g)^2}{\gamma \cdot (1-\tilde{\beta}\rho_g)^2 \cdot (1-\rho_g\Theta) + \Theta \cdot [1-\tilde{\beta}\rho_g(1-\Lambda\gamma)] \cdot (\phi_\pi - \rho_g) \cdot k} > 0, \\
\psi_\mu^g &= \frac{(\phi_\pi - \rho_g) \cdot \tilde{\beta}k}{(1-\tilde{\beta}\rho_g)^2} \cdot \psi_y^g > 0, \\
\psi_\pi^g &= \frac{k}{1-\tilde{\beta}\rho_g} \cdot \psi_y^g > 0,
\end{aligned} \tag{A-20}$$

and for the *monetary shock*:

$$\begin{aligned}
\psi_y^v &= -\frac{\Theta \cdot [1-\tilde{\beta}\rho_v(1-\Lambda\gamma)] \cdot (1-\tilde{\beta}\rho_v)}{\gamma \cdot (1-\tilde{\beta}\rho_v)^2 \cdot (1-\rho_v\Theta) + \Theta \cdot [1-\tilde{\beta}\rho_v(1-\Lambda\gamma)] \cdot (\phi_\pi - \rho_v) \cdot k} < 0, \\
\psi_\mu^v &= \frac{\tilde{\beta}}{1-\tilde{\beta}\rho_v} + \frac{(\phi_\pi - \rho_v) \cdot \tilde{\beta}k}{(1-\tilde{\beta}\rho_v)^2} \cdot \psi_y^v > 0, \\
\psi_\pi^v &= \frac{k}{1-\tilde{\beta}\rho_v} \cdot \psi_y^v < 0.
\end{aligned} \tag{A-21}$$

The multipliers for i.i.d. shocks, as in Section 3.5.4, can be obtained imposing $\rho_g = \rho_v = 0$. The same qualitative implications for the effects of monetary and fiscal shocks (that we have in the i.i.d. case) can be derived in the more general case of persistent shocks. In particular, notice that higher values for Θ imply larger multipliers for output and inflation, in absolute value.

[73] $\mathcal{P}(z)$ is continuous over the interval $[0, 1]$ with $\mathcal{P}(0) > 0$ and $\mathcal{P}(1) < 0$.

B Empirical estimation

B.1 Dataset

The Panel Survey of Income Dynamics (PSID) is a panel survey, directed by the University of Michigan, that gathers data on demographic and economic characteristics of U.S. families as a whole and on individuals within the family. Data were collected annually from 1968 to 1997 and biennially thereafter. The initial sample in 1968 – of about 5,000 families and 18,000 individuals – consisted of a nationally representative sample and an over-sample of low-income families. The PSID sample has grown over time, as children and grandchildren split-off (i.e., move out from the original family and form a new one). In 1997/1999 and 2017/2019, samples of immigrants have been added. Information is collected through in-person, telephone, and computer-assisted interviewing methods.

We focus on the biannual waves from 1969 to 2019. In particular, we use merged Family files (that contain family-level variables for each wave) from 1969 to 2019 and the Individual file (that contains individual-level variables for each wave, from 1968 to 2019, together with family and individual identifiers over time). We download the data through the "cross-year index" tool available at the link `https://simba.isr.umich.edu/DC/i.aspx` (that allows to directly download selected variables and provide Family and Individual files already merged).

Data cleaning. We restrict the PSID sample to family heads, aged between 25 and 65 years old. First, we remove individuals who misreport age across waves (by more than 2 years from the correct age). We also drop individuals who misreport gender across waves and we drop families that experience a change in the head.[74] Second, we exclude from the analysis individuals with top-coded earnings (0.02% of the sample) and low-income observations (i.e., those with less than $1,000 in 2012 dollars),[75] to focus on individuals with more attachment to the labor force. Then, we keep only individuals who have at least two consecutive waves of (non-missing) earnings (in the trimmed sample).[76] Finally, we drop individuals who report outliers of excessive two-years earnings growth or decline (above 500% or below -300% respectively).[77]

Variables definition. The main variables contained in our final dataset are:

- *Earnings* (of the household's head): they include labor part of farm income and business income, wages, bonuses, overtime, commissions, professional practice, labor part of income

[74] Those families that report at least in one wave *change in family composition* different from "No change in family members," "Change in members other than Head or Wife," or "Head same but Wife left/died and/or Head has new Wife."

[75] Out of 124,947 observations (by observation we mean an individual observed at a given point in time), 14.6% are zero-earnings, and 1.02% have non-zero earnings below $1,000 (2012 dollars). We also exclude very high-income individuals, with more than $1,000,000 in 2012 dollars (0.04% of the sample).

[76] They represent 95% of the trimmed sample.

[77] They are 1.94% of the trimmed sample. Overall, our final sample has 109,688 observations. 89.4% of them are used in the analysis (the remaining ones have either top-coded, zero, below $1,000, or above $1,000,000 earnings).

from roomers and boarders or business income. Up to the 1993 wave, earnings include farm income and the labor portion of business income. These are excluded since the 1994 wave. Notice that earnings data in each wave refer to the previous year.

- *Gender* (of the household's head): either male or female.
- *Age* (of the household's head).
- *Change in family composition*: can be "No change in family members," "Change in members other than Head or Wife," "Head same but Wife left/died and/or Head has new Wife."

Each individual is uniquely identified by the *ID* variable. Moreover, the GDP and PCE price index series come from the U.S. Bureau of Economic Analysis.[78] Finally, we deflate all nominal variables using the PCE (set to 100 in 2012). Therefore, real earnings and real GDP are in 2012 dollars.

B.2 Estimators and standard errors

Conditional earnings volatilities estimator. As we described in Section 4.2, to obtain conditional earnings volatilities, first, we estimate with POLS the model of equation (28), using two-years lagged earnings, split into *deciles* (denoted as g^d) and we store the residuals $\epsilon_{i,t}$. Then, for each year t, we compute their sample variance within the lagged earnings bin g^d_{t-1}. Finally, we average these estimates over time. Our conditional variance estimator is, therefore, defined by:

$$Var\left(\epsilon_{i,\cdot}|g^d\right) = \frac{1}{T}\sum_{t=1}^{T}\Bigg(\underbrace{\frac{1}{|S(g^d_{t-1})|}\sum_{i\epsilon S(g^d_{t-1})}\left[\epsilon_{i,t}-\bar{\epsilon}(g^d_{t-1})\right]^2}_{Var\left(\epsilon_{i,t}|g^d_{t-1}\right)\text{: conditional variance in year } t}\Bigg), \tag{B-1}$$

where $S(g^d_{t-1})$ is the set of all individuals who belong to the g^d-th earnings decile in $t-1$, while $\bar{\epsilon}(g^d_{t-1})$ is the average of residuals of individuals in $S(g^d_{t-1})$.[79]

$$S(g^d_{t-1}) := \{i : \text{earnings group of } i \text{ in } t-1 \text{ is } g^d\}$$
$$\bar{\epsilon}(g^d_{t-1}) = \frac{1}{|S(g^d_{t-1})|}\sum_{i\epsilon S(g^d_{t-1})}\epsilon_{i,t}$$

Sufficient statistics. We redefine the sufficient statistics of Proposition 3 to take into account the log specification of the empirical model. The average individual risk effect is the average of the semi-elasticity of conditional earnings volatilities out of lagged earnings and is estimated as an average of individual risk effects ψ_g:

$$\hat{\mathbb{E}}\left(\frac{\partial\sigma^2_{i,t+1}}{\partial y_{i,t}/y_{i,t}}\right) = \frac{1}{5}\sum_{g=1}^{5}\psi_g.$$

[78] They can be downloaded here: GDP, PCE.
[79] The symbol $|\cdot|$ denotes the cardinality of the set.

The average income cyclicality effect is the average of the elasticity of individual earnings out of aggregate output and is estimated as an average of individual income cyclicalities β_g:

$$\hat{\mathbb{E}}\left(\frac{\partial y_{i,t}/y_{i,t}}{\partial y_t/y_t}\right) = \frac{1}{5}\sum_{g=1}^{5}\beta_g.$$

The risk-cyclicality covariance effect is the covariance between the previous two objects:

$$\hat{Cov}\left(\frac{\partial \sigma^2_{i,t+1}}{\partial y_{i,t}/y_{i,t}}, \frac{\partial y_{i,t}/y_{i,t}}{\partial y_t/y_t}\right) = \frac{1}{5}\sum_{g=1}^{5}\psi_g \cdot \beta_g - \left(\frac{1}{5}\sum_{g=1}^{5}\psi_g\right)\cdot\left(\frac{1}{5}\sum_{g=1}^{5}\beta_g\right).$$

Standard errors. We use the central limit theorem to argue that the variance estimator $Var\left(\epsilon_{i,t}|g^d_{t-1}\right)$ (that is an average of the squared residuals) is asymptotically normal. We compute the confidence intervals using the standard errors of the squared residuals (within each earnings bin and averaging across years). Turning to individual risk effects, we treat the variance estimators as uncorrelated across time and acrosss lagged earnings bins. Applying Slutsky theorem, the estimators of the marginal changes in earnings volatility are then asymptotically normal (treating average income deciles as determinisitic).

B.3 Robustness

In addition to our baseline model of (28), – specified in levels with two-years lagged earnings bins, estimated with POLS – we verify the robustness of our results to different specifications and estimation strategies.

First, Figure B-1 shows that our baseline results hold across the following variations:[80] (1) log-earnings and log-GDP (de-trended) are specified in first-difference instead than in levels (i.e., we consider earnings growth and GDP growth, following Broer et al., 2022, Guvenen et al., 2017, and Parker et al., 2010); (2) we group individuals in lagged earnings bins based on four-years lagged earnings instead of two-years; (3) we estimate our model with the fixed effect estimator (to control for individual fixed effects).

Second, keeping our baseline model, we repeat the estimation on different samples where we: (1) remove observations with real earnings that are less than \$100; (2) we do not remove outliers for excessive earnings growth/loss; (3) we remove excessive earnings growth/loss (above 300% and below 100%). Figure B-2 shows that our results hold across these samples.

Finally, we repeat our estimation separately for male/female and for three different age groups (25-34, 35-49, 50-65). This robustness check is useful, given in particular the fact that our sample is composed mainly by male household's heads (around 80% of the sample). Although with larger noise, given the smaller sample sizes, we find estimates similar to those in the baseline model. See Figures B-3 and B-4.

[80] See also Tables B-1, B-2, and B-3.

Bayesian Bootstrap. We estimate our baseline model of (28) (earnings in log-levels and two-years lagged earnings bins) by applying a Bayesian bootstrap procedure. This bootstrap is based on the idea that each individual observation is the realization of a multinomial distribution. The least-squares coefficient is then the minimizer of a weighted sum of squared residuals over the sample space[81] where each element of the sample space is weighted by the probability with which that element is observed in the data (i.e., using the probabilities of the multinomial distribution). By imposing a (non-informative) Dirichlet prior over these probabilities, it is possible to re-express the least-squares coefficient as the minimizer of the sum of squared residuals where each (realized) individual observation is weighted by the square root of a random draw from an exponential distribution (with scale parameter equal to 1).

We implement the bootstrap with the following steps: (1) we take 1,000 random draws from an exponential distribution (with scale parameter equal to 1) for each individual in our sample;[82] (2) we weight each individual by the squared root of the random draw; (3) we compute the (weighted) least-squares coefficients, the estimates for individual risk effects, and for the sufficient statistics of Proposition 3. The output of the algorithm is a set of distributions (i.e., of vectors of 1,000 elements) for individual income cyclicalities, earnings standard deviations (over lagged earnings deciles), individual risk effects, and the sufficient statistics. We then compute the mean, the 5th, and the 95th percentiles of the distributions of these estimates (to obtain 90 percent level credible intervals).[83]

Figure B-5 shows the estimates of the standard deviations, income cyclicalities, and risk effects. We obtain similar results to the baseline estimation strategy, with smaller credible intervals for standard deviations and individual risk effects. Finally, we find significant estimates for our sufficient statistics, similar to those in the baseline model (see Table B-5).

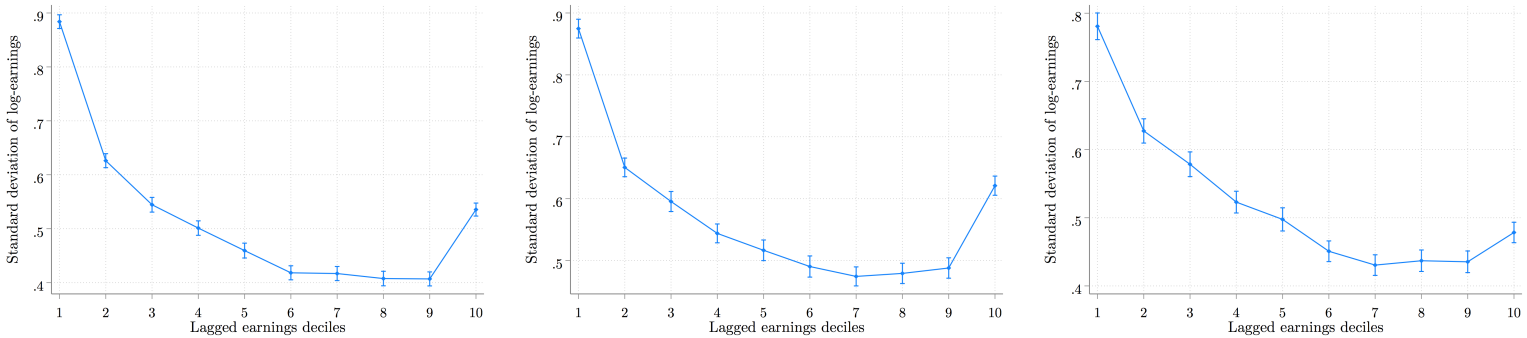
[81] Defined as the set of all non-identical observed individual realizations.
[82] Each agent in the panel has her own random draw, applied for all the years she is in the panel.
[83] The Bayesian name for the confidence interval.

B.4 Figures and tables

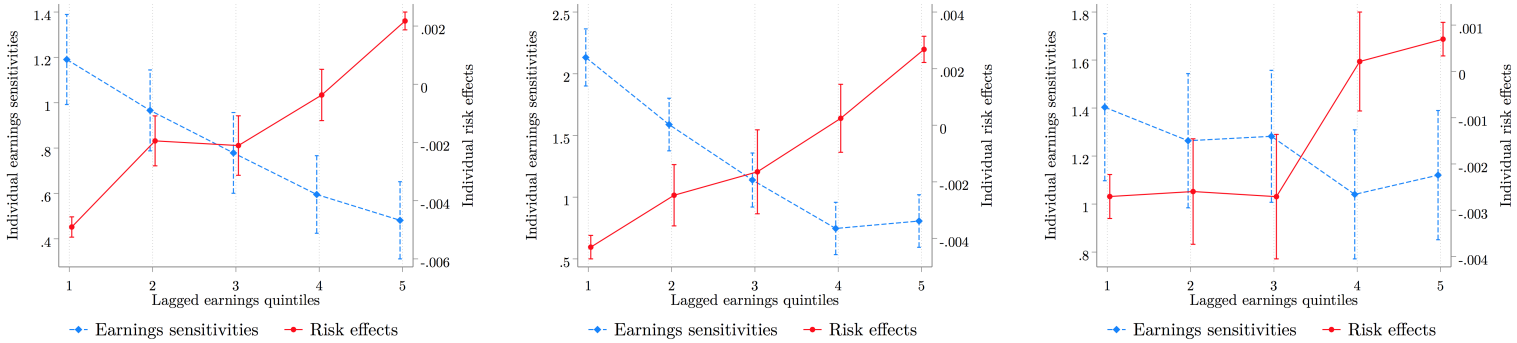
B.4.1 Figures

Figure B-1: Robustness: varying the specification

(a) Estimates of standard deviation of log-earnings



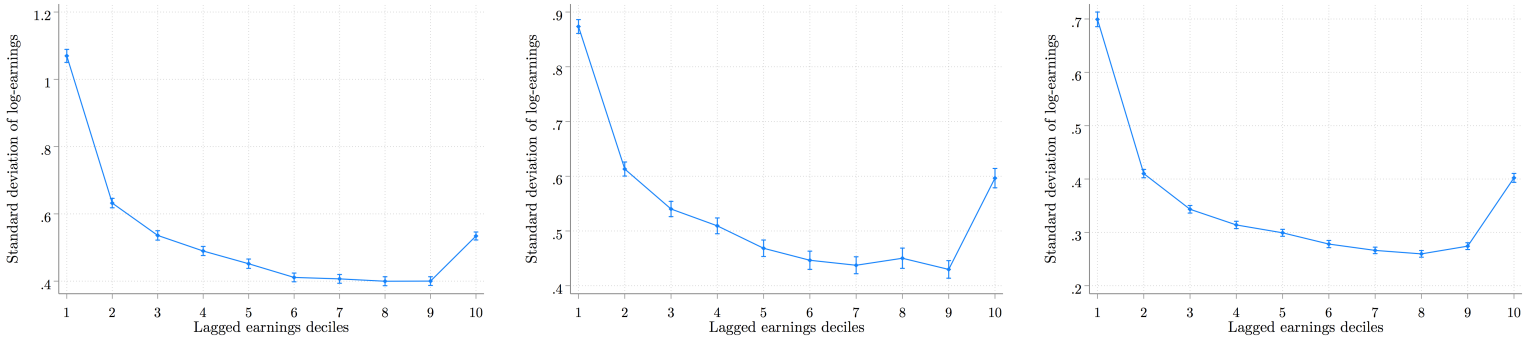
(b) Individual earnings sensitivities and risk effects



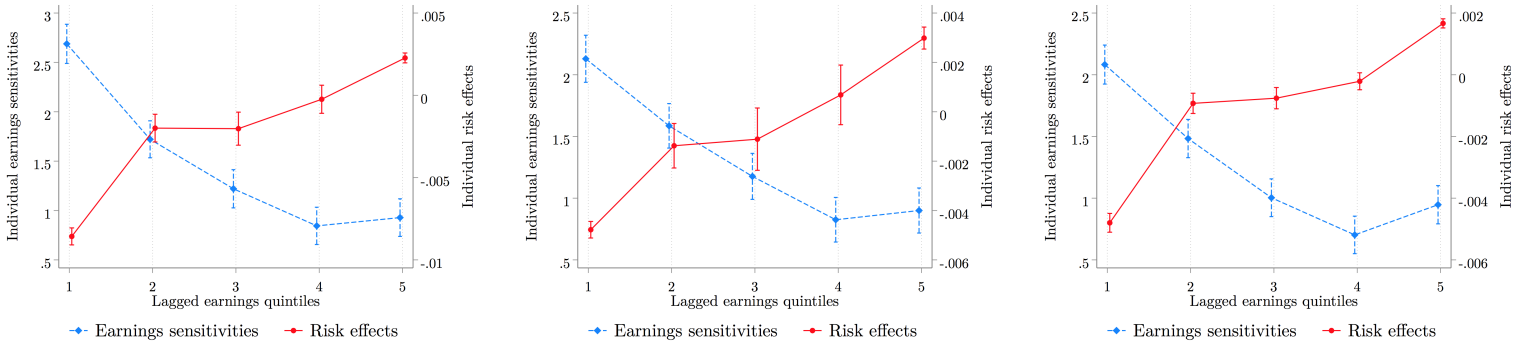
Notes. Panel A shows the standard deviation of the residuals of the log-earnings process for each decile of the lagged earnings distribution. Panel B shows individual earnings sensitivities and risk effects for each quintile of the lagged earnings distribution. The left panel shows the model in levels, with two-years lagged earnings, estimated with fixed effects. The middle panel shows the model in levels with four-years lagged earnings (estimated with POLS). The right panel shows the model in first-difference with four-years lagged earnings (estimated with POLS). Confidence intervals at the 90% level are also shown.

Figure B-2: Robustness: varying the sample

(a) Estimates of standard deviation of log-earnings



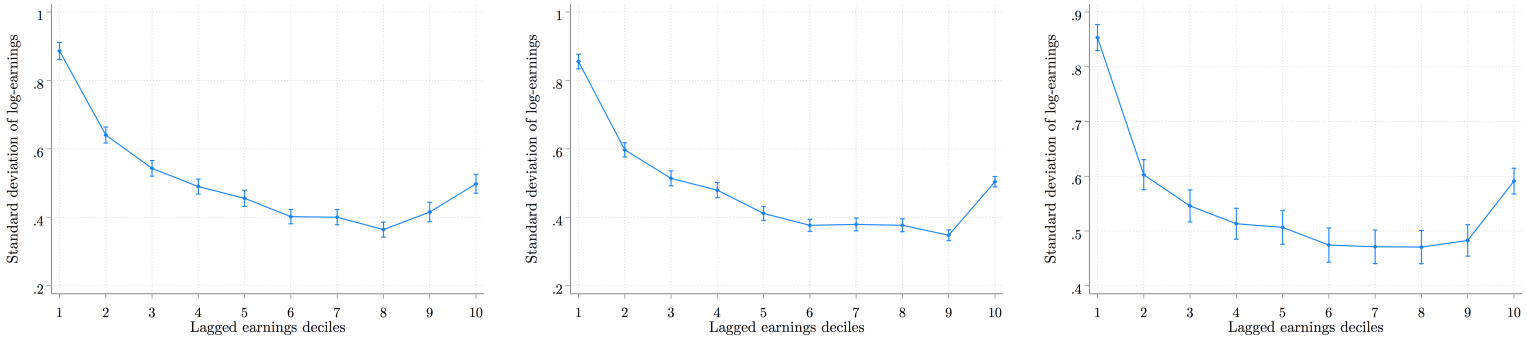
(b) Individual earnings sensitivities and risk effects



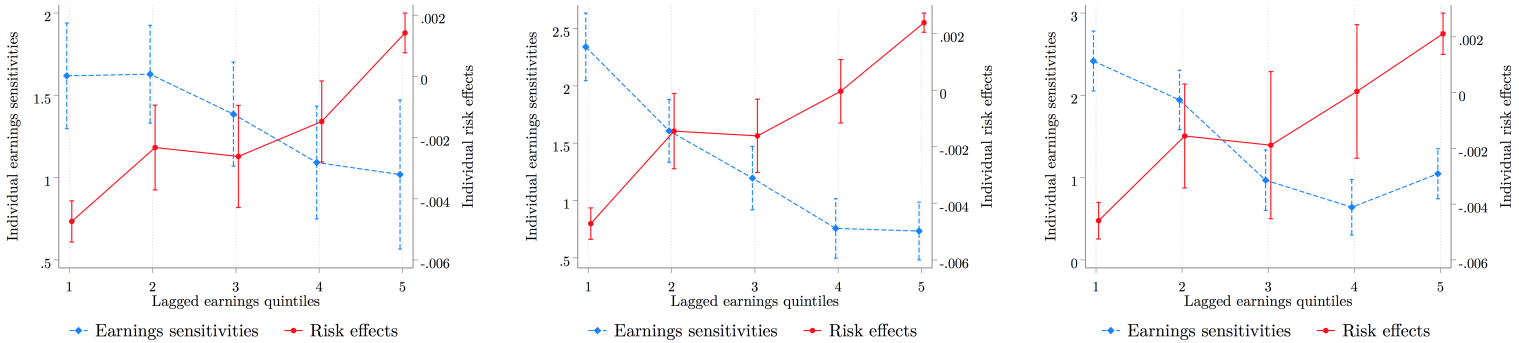
Notes. Panel A shows the standard deviation of the residuals of the log-earnings process for each decile of the (two-years) lagged earnings distribution. Panel B shows individual earnings sensitivities and risk effects for each quintile of the (two-years) lagged earnings distribution. We use log-earnings in levels and the POLS estimator. The left panel uses the sample where we exclude low-income observation with less than $100, the middle panel does not exclude anyone based on excessive earnings growth/loss; the right panel excludes those with two-years excessive earnings growth/loss of more than 300% and 100% respectively. Confidence intervals at the 90% level are also shown.

Figure B-3: Robustness: by age

(a) Estimates of standard deviation of log-earnings



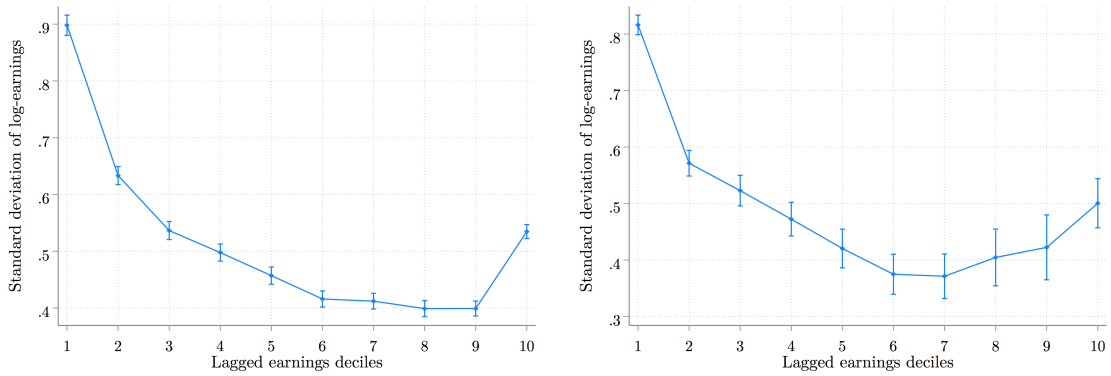
(b) Individual earnings sensitivities and risk effects



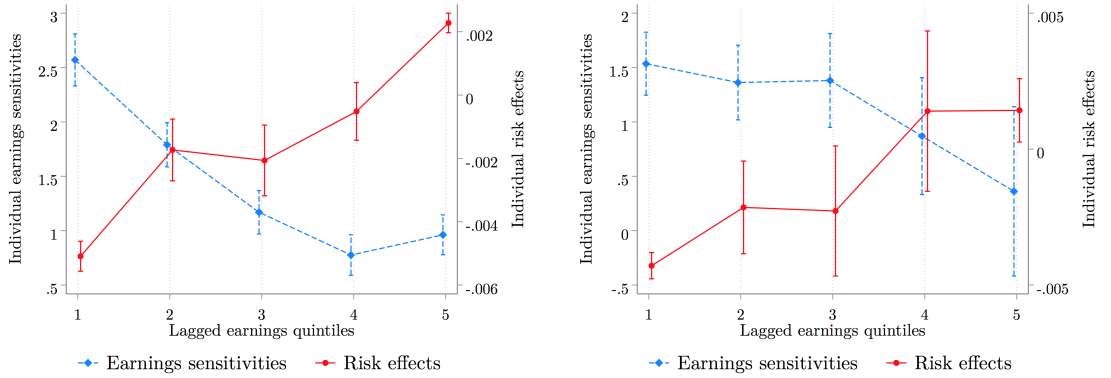
Notes. Panel A shows the standard deviation of the residuals of the log-earnings process for each decile of the (two-years) lagged earnings distribution. Panel B shows individual earnings sensitivities and risk effects for each quintile of the (two-years) lagged earnings distribution. We use log-earnings in levels and the POLS estimator. The left panel refers to estimates for individuals aged 20-34, the middle panel to those aged 35-49, the right panel to those ages 50-65. Confidence intervals at the 90% level are also shown.

Figure B-4: Robustness: by gender

(a) Estimates of standard deviation of log-earnings



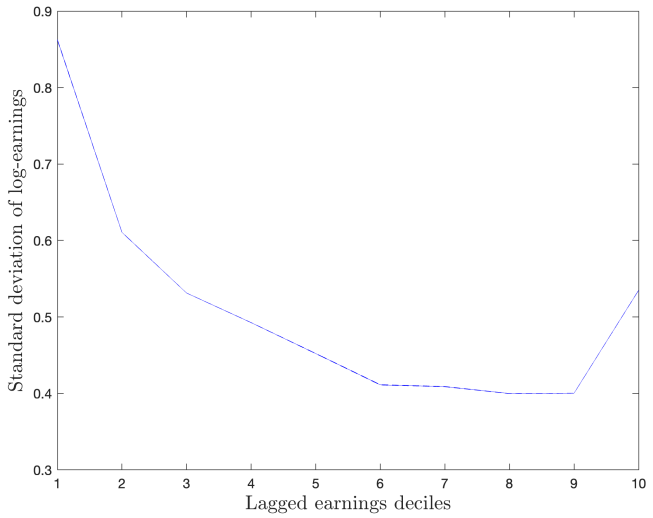
(b) Individual earnings sensitivities and risk effects



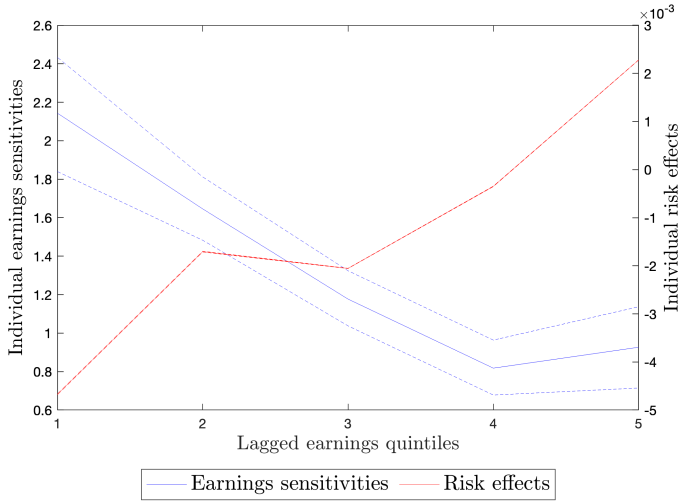
Notes. Panel A shows the standard deviation of the residuals of the log-earnings process for each decile of the (two-years) lagged earnings distribution. Panel B shows individual earnings sensitivities and risk effects for each quintile of the (two-years) lagged earnings distribution. We use log-earnings in levels and the POLS estimator. The left panel refers to estimates for male individuals, while the right panel refers to estimates for female individuals. Confidence intervals at the 90% level are also shown.

Figure B-5: Robustness: Bayesian Bootstrap

(a) Estimates of standard deviation of log-earnings



(b) Individual earnings sensitivities and risk effects



Notes. Panel A shows the standard deviation of the residuals of the log-earnings process for each decile of the (two-years) lagged earnings distribution. Panel B shows individual earnings sensitivities and risk effects for each quintile of the (two-years) lagged earnings distribution. We use log-earnings in levels and apply the Bayesian bootstrap procedure described in Appendix B.3. We report the mean of the empirical distribution of our estimates (solid lines) together with the 5th and 95th percentiles (dashed lines) as the lower and the upper bounds of the 90% level credible intervals.

B.4.2 Tables

Table B-1: Estimation of individual earnings sensitivities - Two-years lagged earnings quintiles

	log-earnings		**log-earnings**		**log-earnings**	
	(1)		(2)		(3)	
quintile 1	9.404***	(0.010)	9.500***	(0.011)	-0.490***	(0.019)
quintile 2	10.030***	(0.010)	10.089***	(0.010)	-0.325***	(0.017)
quintile 3	10.416***	(0.010)	10.453***	(0.010)	-0.185***	(0.016)
quintile 4	10.710***	(0.009)	10.733***	(0.010)	-0.088***	(0.015)
quintile 5	11.089***	(0.009)	11.110***	(0.010)	-	
quintile 1 × trend	0.017***	(0.001)	0.016***	(0.001)	-0.003**	(0.001)
quintile 2 × trend	0.014***	(0.001)	0.014***	(0.001)	-0.002**	(0.001)
quintile 3 × trend	0.011***	(0.001)	0.012***	(0.001)	-0.003**	(0.001)
quintile 4 × trend	0.013***	(0.001)	0.015***	(0.001)	0.001	(0.001)
quintile 5 × trend	0.024***	(0.001)	0.025***	(0.001)	0.009***	(0.001)
Female			-0.142***	(0.005)	-	
age: 35-49			-0.001	(0.005)	0.149***	(0.007)
age: 50-65			-0.098***	(0.005)	0.133***	(0.012)
Individual earnings sensitivities						
quintile 1 × log-GDP	2.232***	[2.046,2.417]	2.146***	[1.961,2.330]	1.191***	[0.992,1.389]
quintile 2 × log-GDP	1.733***	[1.558,1.909]	1.644***	[1.470,1.819]	0.966***	[0.788,1.145]
quintile 3 × log-GDP	1.215***	[1.034,1.397]	1.178***	[0.997,1.358]	0.779***	[0.601,0.957]
quintile 4 × log-GDP	0.788***	[0.611,0.965]	0.818***	[0.643,0.994]	0.595***	[0.424,0.766]
quintile 5 × log-GDP	0.933***	[0.754,1.112]	0.927***	[0.749,1.104]	0.481***	[0.311,0.652]
Observations	81655		81655		81655	

Notes. The dependent variables are log-earnings in levels. These are regressed on indicators for (two-years) lagged earnings quintiles, also interacted with a linear time trend, and with (de-trended) log-GDP. Columns 2 and 3 add indicator variables for female and age groups (35-49 and 50-65, while 25-34 is omitted). Columns 1 and 2 estimate the model with POLS, while column 3 uses a fixed effects estimator. All variables are defined in Appendix B.1. We report 90% confidence intervals for individual earnings sensitivities. Otherwise, we show standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B-2: Estimation of individual earnings sensitivities - Four-year lagged earnings quintiles

	log-earnings		**log-earnings**		**log-earnings**		**log-earnings growth**	
	(1)		(2)		(3)		(4)	
quintile 1	9.469***	(0.013)	9.621***	(0.014)	-0.265***	(0.024)	0.142***	(0.006)
quintile 2	10.018***	(0.013)	10.121***	(0.013)	-0.184***	(0.021)	0.074***	(0.006)
quintile 3	10.403***	(0.012)	10.478***	(0.013)	-0.086***	(0.019)	0.060***	(0.006)
quintile 4	10.712***	(0.012)	10.769***	(0.012)	-0.014	(0.017)	0.054***	(0.006)
quintile 5	11.043***	(0.012)	11.107***	(0.013)	-		0.046***	(0.006)
quintile 1 × trend	0.020***	(0.001)	0.019***	(0.001)	-0.005***	(0.002)		
quintile 2 × trend	0.016***	(0.001)	0.017***	(0.001)	-0.008***	(0.001)		
quintile 3 × trend	0.013***	(0.001)	0.014***	(0.001)	-0.009***	(0.001)		
quintile 4 × trend	0.013***	(0.001)	0.015***	(0.001)	-0.009***	(0.001)		
quintile 5 × trend	0.025***	(0.001)	0.026***	(0.001)	-0.002	(0.001)		
Female			-0.158***	(0.006)	-		-0.005	(0.006)
age: 35-49			-0.040***	(0.006)	0.172***	(0.007)	-0.041***	(0.005)
age: 50-65			-0.176***	(0.006)	0.185***	(0.014)	-0.127***	(0.006)
Individual earnings sensitivities								
quintile 1 × log-GDP	2.210***	[1.976,2.443]	2.132***	[1.901,2.363]	0.977***	[0.744,1.210]		
quintile 2 × log-GDP	1.681***	[1.465,1.896]	1.588***	[1.375,1.802]	0.839***	[0.635,1.043]		
quintile 3 × log-GDP	1.164***	[0.942,1.385]	1.139***	[0.920,1.358]	0.736***	[0.532,0.941]		
quintile 4 × log-GDP	0.685***	[0.471,0.899]	0.746***	[0.535,0.958]	0.637***	[0.443,0.832]		
quintile 5 × log-GDP	0.801***	[0.586,1.016]	0.807***	[0.594,1.020]	0.441***	[0.248,0.634]		
quintile 1 × Δ log-GDP							1.101***	[0.807,1.396]
quintile 2 × Δ log-GDP							1.031***	[0.762,1.299]
quintile 3 × Δ log-GDP							1.081***	[0.815,1.346]
quintile 4 × Δ log-GDP							0.914***	[0.655,1.172]
quintile 5 × Δ log-GDP							0.908***	[0.649,1.167]
Observations	66837		66837		66837		65161	

Notes. The dependent variables are log-earnings in levels (columns 1, 2, 3) and two-years log-earnings growth (column 4). These are regressed on indicators for (four-year) lagged earnings quintiles, also interacted with a linear time trend, and with (de-trended) log-GDP. Columns 2, 3, and 4 add indicator variables for female and age groups (35-49 and 50-65, while 25-34 is omitted). Columns 1, 2, and 4 estimate the model with POLS, while column 3 uses a fixed effects estimator. All variables are defined in Appendix B.1. We report 90% confidence intervals for individual earnings sensitivities. Otherwise, we show standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B-3: Estimation of individual earnings sensitivities - Lagged earnings deciles

	log-earnings	
	(1)	
decile 1 × log-GDP	2.350***	[2.093,2.606]
decile 2 × log-GDP	2.009***	[1.765,2.253]
decile 3 × log-GDP	1.692***	[1.448,1.937]
decile 4 × log-GDP	1.562***	[1.332,1.792]
decile 5 × log-GDP	1.485***	[1.236,1.734]
decile 6 × log-GDP	0.865***	[0.624,1.107]
decile 7 × log-GDP	0.840***	[0.602,1.078]
decile 8 × log-GDP	0.800***	[0.561,1.038]
decile 9 × log-GDP	0.959***	[0.719,1.200]
decile 10 × log-GDP	0.945***	[0.704,1.186]
Observations	81655	

Notes. The dependent variables are log-earnings in levels. These are regressed on indicators for (two-years) lagged earnings deciles, also interacted with a linear time trend, and with (de-trended) log-GDP. Indicator variables for female and age groups (35-49 and 50-65, while 25-34 is omitted) are also included. We use the POLS estimator. The table shows only individual earnings sensitivities. All variables are defined in Appendix B.1. We report 90% confidence intervals for individual earnings sensitivities. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B-4: Sufficient statistics: varying the specification

Statistic	**Estimate**		
	(1)	(2)	(3)
Average individual risk effect	-.143	-.11	-.142
Average individual earnings sensitivities	.802	1.283	1.222
Risk-cyclicality covariance effect	-.056	-.112	-.017
Parameter Γ	-.171	-.253	-.191

Notes. The table shows our estimates of the sufficient statistics in Proposition 3. Column (1) refers to the model in levels with two-years lagged earnings bins, estimated with fixed effects. Column (2) refers to the model in levels with four-year lagged earnings bins, estimated with POLS. Column (3) refers to the model in first-difference with four-year lagged earnings bins, estimated with POLS.

Table B-5: Sufficient statistics: Bayesian Bootstrap

Statistic	**Estimate**	
	(1)	90% Credible Intervals
Average individual risk effect	-.1301	[-.1304,-.1297]
Average individual earnings sensitivities	1.3421	[1.2597,1.4240]
Risk-cyclicality covariance effect	-.0938	[-.1179, -.0681]
Parameter Γ	-.2683	[-.2980,-.2363]

Notes. The table shows our estimates of the sufficient statistics in Proposition 3. We estimate the model in levels with two-years lagged earnings bins applying the Bayesian bootstrap procedure described in Appendix B.3. We report the mean of the empirical distribution of our estimates (column 1) together with the 5th and 95th percentiles as the lower and upper bounds of the credible intervals.

C Model with HtM agents

Our baseline specification suffers from its inability to match aggregate (annual) MPCs with estimates from microdata. In fact, our CARA utility model does not have borrowing constraints and this generates MPCs of a permanent-income consumer (close to zero). Recall that MPCs out of income are defined as:

$$MPC_{i,t} := \frac{\partial c_{i,t}}{\partial y_{i,t}} = \mu_t - \frac{1}{1+\mu_{t+1}(1+r_t)} \frac{\gamma \mu_{t+1}^2}{2} \frac{\partial \sigma_{i,t+1}^2}{\partial y_{i,t}},$$

and, in the steady state, the previous formula becomes:

$$MPC_i := \frac{\partial c_{i,t}}{\partial y_{i,t}} = \frac{r}{1+r} - \frac{1}{(1+r)^3} \frac{\gamma r^2}{2} \frac{\partial \sigma_{i,t+1}^2}{\partial y_{i,t}} \approx \frac{r}{1+r},$$

where the last approximation holds well for reasonable values of the model parameters.

On the contrary, typical MPC estimates fall between 0.2 and 0.6.[84] A simple way to capture empirically reasonable estimates for the aggregate MPC is to introduce hand-to-mouth agents in our model. Hand-to-mouths (HtM) are agents who can consume, in every period, only their disposable income and who cannot access the financial market. Thus, they have an MPC out of transitory income shocks equal to unity.

C.1 Households

Let us consider the same framework as our baseline model of Section 3, with the only difference that there is a measurable set $\mathcal{H}$ of households of measure $\lambda \in [0,1]$ that are HtM, and a set $\mathcal{U}$ of measure $1-\lambda$ of non-HtM unconstrained agents. We recover the model of Section 3 imposing $\lambda = 0$.

C.1.1 Non-HtM unconstrained households

Every household $i \in \mathcal{U}$ (we denote them by the superscript U) solves the optimization problem described in Section 3.1. The consumption function is:

$$c_{i,t}^U = c_{i,t+1}^U - \frac{1}{\gamma} \log[\beta(1+r_t)] - \frac{\gamma \mu_{t+1}^2}{2} \sigma_{i,t+1}^2 - \mu_{t+1} \epsilon_{i,t+1}^U, \tag{C-1}$$

where μ_t, the MPC out of wealth, follows the recursion:

$$\mu_t = \frac{\mu_{t+1}(1+r_t)}{1+\mu_{t+1}(1+r_t)}, \tag{C-2}$$

as stated in Proposition 1.

[84] For instance, Fagereng et al. (2021) estimate an MPC of 0.55, Kaplan et al. (2014) estimate MPCs between 0.1 and 0.3 (for non-HtM and HtM agents, respectively), and Jappelli and Pistaferri (2014) estimate an MPC of 0.48. See also Shapiro and Slemrod (2003), Johnson et al. (2006), Parker et al. (2013), Fuster et al. (2021).

C.1.2 HtM households

Every HtM agent $i \in \mathcal{H}$ (we denote them by the superscript H) can consume her total income in every period. Her problem can be reformulated as a static one and, given the strict monotonicity of the utility function, the HtM optimal consumption function is simply:

$$c_{i,t}^H = y_{i,t}^H. \tag{C-3}$$

C.2 Aggregation and general equilibrium

Let us assume, in the spirit of Bilbiie (2020), that HtM agents' average consumption has a sensitivity to aggregate consumption equal to $\chi \geq 0$:

$$c_t^H := \frac{1}{\lambda} \int_{\mathcal{H}} c_{i,t}^H di = \chi \cdot c_t. \tag{C-4}$$

The parameter χ captures how average HtM consumption changes out of aggregate consumption.[85] For instance, $\chi > 1$ means that HtM households increase their consumption more than one-to-one with aggregate one.[86]

We compute unconstrained agents average consumption by integrating equation (C-1) with respect to the Lebesgue measure, $c_t^U := \frac{1}{1-\lambda} \int_{\mathcal{U}} c_{i,t}^U di$ and we get:

$$c_t^U = c_{t+1}^U - \frac{1}{\gamma} \log[\beta(1 + r_t)] - \frac{\gamma \mu_{t+1}^2}{2} \frac{1}{1-\lambda} \int_{\mathcal{U}} \sigma_{i,t+1}^2 di. \tag{C-5}$$

The goods market clearing can be reformulated as:

$$y_t = c_t + g_t, \tag{C-6a}$$

$$c_t = \lambda \cdot c_t^H + (1 - \lambda) \cdot c_t^U, \tag{C-6b}$$

where c_t denotes aggregate consumption. Combining (C-4) and (C-6b), we find:

$$c_t^U = \frac{1 - \lambda \cdot \chi}{1 - \lambda} \cdot c_t. \tag{C-7}$$

Finally, substituting the previous expression in (C-5) and using the goods market clearing, we obtain the aggregate AD equation.

$$y_t = y_{t+1} - \Xi \cdot \frac{1}{\gamma} \log[\beta(1 + r_t)] - \Xi \cdot \frac{\gamma \mu_{t+1}^2}{2} \frac{1}{1-\lambda} \int_{\mathcal{U}} \sigma_{i,t+1}^2 di + g_t - g_{t+1} \tag{C-8}$$

[85] This can be equivalently reformulated as the sensitivity of HtM average disposable income to aggregate after-tax income.

[86] For instance, if the government imposes a transfer scheme to redistribute income from unconstrained to HtM agents.

with $\Xi := \frac{1-\lambda}{1-\lambda\cdot\chi}$. Note that if $\chi > 1$ the effects of changes in the real interest rate and of precautionary savings are amplified since $\Xi > 1$.[87] Intuitively, if $\chi > 1$, an increase in aggregate consumption determines a more than proportional increase in HtM average income. Since HtM agents have an MPC equal to 1, their consumption is extremely sensitive to an increase in their income. Therefore, any economic shocks that redistribute earnings relatively more to HtM than to unconstrained households has a larger impact on aggregate output.

C.2.1 Log-linear representation

It is possible to log-linearize equation (C-8) and obtain:

$$\hat{y}_t = \Theta_{HtM} \cdot \left[\hat{y}_{t+1} - \Xi \cdot \frac{1}{\gamma} \cdot \hat{r}_t - \Xi \cdot \Lambda_{HtM} \cdot \hat{\mu}_{t+1} + \left(\hat{g}_t - \hat{g}_{t+1}\right) \right], \tag{C-9}$$

where:

$$\begin{aligned}
\bar{\sigma}^2_{HtM} &:= \frac{1}{1-\lambda} \cdot \int_{[0,1]} \sigma_i^2 di, \\
\Theta_{HtM} &:= \left(1 + \Xi \cdot \frac{\gamma\mu^2}{2}\Gamma\right)^{-1}, \\
\Lambda_{HtM} &:= \gamma \cdot \mu^2 \cdot \bar{\sigma}^2_{HtM}.
\end{aligned}$$

[87] We need to impose that $\lambda < \chi^{-1}$.